INTRACRANIAL-HEMORRHAGE-DETECTION

Detecting Intracranial Haemorrhage on Noncontract CT scans using ML and Deep Learning Models

Abstract

Automated detection of intracranial hemorrhage (ICH) on non-contrast CT scans is vital for quick diagnosis in emergency care. However, challenges like small or subtle lesions, varied imaging protocols, and differences between medical centers hinder the development of effective automated solutions. This thesis evaluates three methods for slice-level ICH detection using the multi-center CQ500 dataset. First being,**Traditional machine-learning classifiers** that rely on radiomic features like intensity, shape, and texture.Then **Convolutional neural networks (CNNs)** trained from scratch or using transfer learning from models like ImageNet, MobileNetV2, EfficientNet-B0, and ResNet-50.Finally, **LightGBM models** that also use the same set of radiomic features.

To improve the safety and reliability of these models, we used techniques like Monte Carlo Dropout and selective prediction to measure predictive uncertainty. This allowed us to automatically triage high-confidence cases while flagging ambiguous ones for expert review. Our results showed that models based on handcrafted features and transfer-learning CNNs had only modest performance (AUC ~0.55–0.67), and networks trained from scratch generalized poorly. In contrast, using uncertainty-guided triage significantly improved dependability, proving that calibrated uncertainty can make modest models safer and more practical for use in settings where training data and volumetric context are limited. We conclude with recommendations for future research, including self-supervised learning, 2.5D/3D modeling, and multi-site validation, to create clinically trustworthy AI in neuro-emergency imaging.

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CHAPTER 1- INTRODUCTION

Intracranial hemorrhage (ICH) is a leading cause of death and long-term disability related to stroke. Rapid and accurate identification on non-contrast CT scans is crucial for treatment, but interpretation can be difficult due to subtle bleeding, complex anatomy, and variations in CT scanners and acquisition protocols. In busy emergency rooms, radiologists face heavy workloads, making computer-assisted detection a promising way to speed up triage and ensure consistency. However, deploying these systems is complicated by differences in data distribution, labeling inconsistencies, and the risk of automation errors that could endanger patients.

Traditional radiomic methods, which use features like intensity, texture, and shape, are efficient and easy to interpret, but their performance often falls short when lesions are subtle or when data comes from various medical centers. Deep convolutional neural networks (CNNs) have shown promise on well-curated datasets, but training them from scratch on limited CT volumes can lead to overfitting. Transfer learning from models trained on natural images may also fail to capture the specific contrast and texture patterns of CT scans. These limitations highlight the need for a systematic comparison of existing methods, with a strong focus on predictive uncertainty as a safety measure for clinical use.

This thesis evaluates three main types of models for slice-level ICH detection using the CQ500 dataset: traditional machine-learning classifiers, various CNNs (including MobileNetV2, EfficientNet-B0, and ResNet-50), and LightGBM models. To boost reliability, we use Monte Carlo Dropout to estimate predictive confidence and apply selective prediction to flag low-confidence slices for a human radiologist to review. This human-AI collaboration aims to provide dependable assistance by automating clear-cut cases and leaving complex ones for human experts.

The main contributions of this work are:

- A unified framework for preprocessing and evaluating the CQ500 dataset, including DICOM conversion, normalization, and quality control.
- A direct comparison of radiomic models, CNNs, and gradient boosting methods under consistent conditions.
- The integration of uncertainty quantification and selective deferral to enhance safety in automated triage.
- An analysis of the challenges to model generalization, such as overfitting and the lack of 3D context, and their implications for future research.

Our findings show that while radiomic and transfer-learning models achieve only moderate performance, and scratch-trained CNNs generalize poorly, a key finding is that selective prediction based on uncertainty significantly improves reliability. This emphasizes the critical role of uncertainty modeling in deploying AI for emergency neuroimaging and motivates further research into domain-specific pretraining, volumetric modeling, and multi-site validation.

The rest of the thesis is organized as follows:

- Chapter 2: Review of past work on computer-assisted brain imaging, especially for ICH.
- Chapter 3: Methods used for all three approaches, including data preparation and model building.
- Chapter 4: Results and comparisons of the three approaches.
- Chapter 5: Discussion of what the results mean for clinical practice, study limitations, and future directions.
- Chapter 6: Summary of contributions and recommendations for clinical use.

This research aims to show how different computer-based methods can support faster and more accurate diagnosis of brain bleeds, helping doctors make better decisions and improve patient outcomes.

CHAPTER 2- LITERATURE REVIEW

1. **Deep learning algorithms for detection of critical findings in head CT scans: a retrospective study.**

Chilamkurthy [1] (2018) carried out a landmark multi-centre retrospective study in which they developed deep learning models to detect nine urgent findings on non-contrast head CT scans. These included intracranial haemorrhage (ICH) and its subtypes, calvarial fractures, midline shift, and mass effect. The training dataset was exceptionally large, comprising more than 313,000 scans, and the models were validated on two separate cohorts—Qure25k and CQ500—to demonstrate both generalisability and clinical relevance. Each clinical target was framed as a scan-level classification problem, with the models producing continuous confidence scores. Performance was assessed using ROC analysis at both high-sensitivity and high-specificity operating points, reflecting the practical trade-offs needed in emergency triage, where speed, accuracy, and manageable false-alert rates are all crucial.
For internal validation on Qure25k (21,095 scans), labels were derived from radiology reports using a validated NLP pipeline. The models achieved strong discrimination across most tasks, with AUCs of 0.92 for ICH, 0.92 for calvarial fracture, and 0.93 for midline shift. Performance for mass effect was lower at 0.86, suggesting that certain findings are more challenging due to their varied imaging appearance. External validation on CQ500 (491 scans, with ground truth based on consensus from three experienced radiologists) confirmed robust generalisation. Reported AUCs were 0.94 for ICH, 0.96 for calvarial fracture, 0.97 for midline shift, and 0.92 for mass effect. Subtype-specific ICH detection was also strong, with AUCs of 0.97 for extradural and 0.96 for subarachnoid haemorrhage. These results highlighted the system's potential for clinically meaningful triage across institutions and scanner types.
The study also examined the reliability of the ground truth in CQ500, finding excellent inter-rater agreement for ICH and intraparenchymal haemorrhage (Fleiss' κ around 0.77–

0.78). Agreement was only fair to moderate for subdural haemorrhage and calvarial fractures, pointing to the inherent ambiguity of these findings and the label noise that can limit model performance. When the algorithms were compared directly with individual radiologists at a high-sensitivity operating point, sensitivities were similar, but the models generally showed lower specificity. This imbalance raises concern for false-positive alerts, which, if not carefully managed, could contribute to alarm fatigue in clinical practice.

Despite its impact, the study also had important limitations. Predictive uncertainty was not quantified or calibrated, meaning that the models' confidence scores were not interpretable as true probabilities and lacked mechanisms for selectively deferring borderline or out-of-distribution cases. Error analysis further showed difficulties in detecting very small intraventricular haemorrhages and findings in the skull-base region. In addition, the use of report-derived labels for Qure25k—while validated—still risked inheriting biases from free-text documentation. These gaps highlight the need for future work to integrate uncertainty quantification methods, such as Monte Carlo Dropout or ensemble models with temperature scaling, and to provide localisation or slice-level uncertainty maps. Such improvements could help transform strong AUC performance into trustworthy, workflow-compatible decision support tools for acute neuroimaging triage.

2. **Impact of a deep learning-based brain CT interpretation algorithm on clinical decision-making for intracranial hemorrhage in the emergency department**

Choi [2](2024) examined how a deep learning–based intracranial hemorrhage detection system (DLHD) influences real-time decision-making in the emergency department (ED). In a prospective simulation, ten ED clinicians first interpreted the same set of non-contrast head CT scans without assistance and then repeated the review with the algorithm's support. The DLHD system, approved for clinical use by the Korean FDA, was built as an ensemble of five 2D U-Net–Inception models designed to segment lesions, detect hemorrhage subtypes such as subdural and subarachnoid bleeds, and identify small lesions. Unlike earlier research prototypes, the study evaluated a clinically deployable tool.

In retrospective testing on 2,146 CT scans from a single site, the algorithm achieved a sensitivity of 70.81% (95% CI 66.65–74.97), specificity of 86.72% (85.10–88.34), accuracy of 83.32% (81.74–84.90), and an AUROC of 0.788 (0.765–0.810). For the prospective simulation, the authors selected 111 cases balanced across true positive, true negative, false positive, and false negative categories in order to stress-test the system under different clinical scenarios. With DLHD support, inexperienced clinicians improved their sensitivity from 59.33% to 72.67% ($p < 0.001$), but their specificity dropped from 65.49% to 53.73% ($p = 0.001$). Their decision consistency, measured by Cohen's κ, remained moderate at around 0.42–0.43. By contrast, experienced clinicians showed no significant differences in sensitivity, specificity, accuracy, or AUROC with assistance, and maintained higher agreement levels ($\kappa \approx 0.74$–0.77). These findings suggest that expertise helps clinicians filter algorithmic outputs and avoid overreliance during triage.

The study also explored common sources of algorithmic error and their influence on clinical decisions. DLHD sometimes misclassified tumors, calcifications, and skull-base hyperdensities as hemorrhage. These errors led to false-positive–driven decision changes in nearly 45% of cases for less experienced readers, compared with about 18% for more experienced readers. Such results mirror broader reports of AI deployment in high-volume, low-prevalence settings, where excessive false alerts can slow interpretation and reduce overall efficiency. The authors highlighted the importance of calibrated thresholds, selective triage strategies, and workflow safeguards to limit alarm fatigue and automation bias.

While the simulation included realistic metadata and a washout period to reduce recall bias, it still represented a controlled environment. The authors emphasized that further studies covering a wider range of pathologies, involving multiple institutions, and extending into real-world clinical trials are needed to validate the system's net clinical benefit. For safer integration, they recommended positioning DLHD as a triage support tool, especially for junior clinicians, while pairing its outputs with calibrated confidence levels, explicit uncertainty displays, and options for deferral when predictions are borderline. They also proposed tailoring alert thresholds and user interfaces to clinician experience, possibly using higher thresholds, uncertainty gating, or conformal prediction techniques to reduce overcalling without sacrificing the system's support value.

Overall, the study provides rare evidence focused directly on clinician interaction with ICH AI tools. It demonstrates that the impact of DLHD depends heavily on reader expertise and deployment design. Choi et al. argue that uncertainty quantification, calibration, and workflow safeguards are essential if algorithmic accuracy is to be translated into reliable and scalable decision support in the ED.

3. **A deep learning algorithm for automatic detection and classification of acute intracranial hemorrhages in head CT scans**

Wang [3] proposed a multi-stage deep learning framework that emulates the diagnostic workflow of radiologists for detecting acute intracranial hemorrhage (ICH) and classifying its five subtypes on non-contrast head CT scans. Their approach integrates a slice-level two-dimensional convolutional neural network (CNN) with two bi-directional gated recurrent unit (GRU) networks to aggregate information across slices. This design introduces three-dimensional contextual awareness and enables adaptive refinement of predictions, effectively capturing dependencies between adjacent slices. Preprocessing mirrored radiological practice by applying standard clinical window settings—brain, subdural, and bone—and stacking the resulting images as separate channels.

The system was primarily trained and evaluated on the RSNA 2019 Brain CT Hemorrhage dataset, which contains more than 25,000 scans. On this benchmark, the ensemble achieved excellent discrimination, with an overall ICH AUC of 0.988 and subtype-level AUCs of 0.996 for intraventricular hemorrhage, 0.992 for intraparenchymal hemorrhage, 0.985 for subarachnoid hemorrhage, 0.983 for subdural hemorrhage, and 0.984 for epidural hemorrhage. At clinically selected operating points, both sensitivity and specificity exceeded 0.94 for most tasks, demonstrating radiologist-level slice-wise performance within the curated dataset.

Robust external generalization was also demonstrated. On the PhysioNet-ICH dataset (75 scans), the model achieved an AUC of 0.964, while on the CQ500 dataset (491 scans), it reached 0.949. Subtype classification performance on CQ500 remained strong, though detection of subarachnoid and subdural hemorrhage showed moderate degradation compared with RSNA results. This variability is consistent with the known challenges of lesion subtlety, anatomical diversity, and protocol heterogeneity across imaging sites. Comparative analyses confirmed that the system delivered superior or comparable scan-level AUCs relative to prior CQ500 benchmarks. Furthermore, Grad-CAM visualizations demonstrated close alignment with radiologist annotations, highlighting hemorrhagic regions and enhancing the model's clinical plausibility.

Methodologically, the two-stage aggregation process represents an important advance over conventional 2D classifiers. By modeling long-range dependencies across slices, the GRU layers not only reduced misclassifications from the upstream CNN but also allowed integration of metadata such as slice thickness. Nevertheless, the framework has limitations. It does not incorporate explicit uncertainty quantification or calibration, reducing interpretability in borderline cases. External evaluations also revealed domain shift effects arising from scanner and protocol differences, while the retrospective design did not include prospective or user-in-the-loop testing. In addition, reliance on curated RSNA labels and the relatively high training complexity may limit direct clinical deployment without further multi-center validation. Although inference speed was practical (~90 ms per slice), these implementation challenges remain relevant for real-world translation.

Taken together, the study by Wang et al. sets a high benchmark for ICH detection and subtype classification by demonstrating the effectiveness of sequence-aware modeling combined with interpretable saliency visualizations. At the same time, the findings highlight the importance of uncertainty calibration and deferral mechanisms to ensure that strong AUC values translate into safe and reliable support for time-critical emergency workflows.

4. **Quantifying Uncertainty in Deep Learning of Radiologic Images**

Faghani [4](2023) argue that deep learning systems in radiology must move beyond deterministic predictions and provide explicit measures of uncertainty, analogous to a radiologist's confidence in their interpretation. They distinguish between two principal forms of uncertainty: **aleatoric**, which stems from inherent noise, variability, or ambiguity within medical images and cannot be eliminated; and **epistemic**, which arises from limited, biased, or incomplete training data and can be reduced through improved modeling or more diverse datasets. By classifying uncertainty into these categories, the authors emphasize that appropriate mitigation depends on its source—curation and preprocessing for aleatoric uncertainty, and expansion of training coverage for epistemic uncertainty.

The authors also stress the importance of differentiating between calibration and uncertainty quantification. While calibration ensures that predicted probabilities match long-term outcome frequencies, it does not guarantee that those probabilities remain trustworthy when a model is faced with unfamiliar or out-of-distribution cases. For this reason, they advocate presenting uncertainty estimates through scalar scores, confidence intervals, and pixel- or region-level uncertainty maps, particularly in segmentation tasks. Such outputs allow clinicians to identify cases where predictions should be treated cautiously and subject to further review rather than automated decision-making.

To operationalize uncertainty quantification, Faghani et al. describe three principal methodological approaches suitable for medical imaging. **Deep ensembles** train multiple independent models and measure the variation in their outputs, providing a robust though computationally expensive estimate of predictive reliability. **Bayesian approximations**, most notably Monte Carlo Dropout, retain dropout layers during inference and generate predictive distributions through repeated stochastic forward passes. **Evidential deep learning** extends the output space of neural networks to include evidence parameters, allowing uncertainty estimation in a single forward pass. Each technique involves trade-offs between computational burden, implementation complexity, and fidelity of the resulting uncertainty measures.

Clinical examples illustrate the practical benefits of uncertainty-aware modeling. In segmentation tasks, uncertainty maps can highlight regions most likely to require expert correction. In classification, separating certain from uncertain cases enables selective deferral, focusing expert attention on ambiguous scans and reducing unnecessary workload. Uncertainty-guided active learning strategies also accelerate model development by directing annotation resources toward the most informative cases. Moreover, some methods improve calibration when models encounter distributional shift, further increasing reliability at clinically relevant operating points.

Despite these benefits, the review identifies several barriers to routine adoption of uncertainty quantification in medical imaging. There is a lack of benchmark datasets that incorporate controlled uncertainty annotations, making it difficult to compare methods. Clinically accessible software tools that integrate uncertainty into existing AI pipelines remain limited, and standardized protocols for mapping uncertainty values onto actionable clinical thresholds are underdeveloped. To advance the field, the authors call for the development of public datasets with engineered uncertainty characteristics, standardized evaluation frameworks that include both calibration and safety-oriented metrics, and open-source libraries that enable per-prediction uncertainty estimation, calibration checking, and deferral strategies.

These recommendations directly inform the present thesis, which incorporates Monte Carlo Dropout and calibration techniques into CNN-based intracranial hemorrhage detection. By generating slice-level uncertainty heatmaps alongside calibrated confidence estimates, this approach aims to provide clinicians with interpretable, trustworthy decision support that aligns with real-world diagnostic needs.

5. **A review of uncertainty quantification in medical image analysis: probabilistic and non-probabilistic methods**

Huang [5] (2024) provide a detailed review of methods for measuring uncertainty in medical image analysis, stressing that this step is essential before machine learning systems can be safely used in clinical practice. They explain the difference between two types of uncertainty. **Aleatory uncertainty** comes from natural variation in the data, such as noise in the scan or patient movement, and cannot be removed. **Epistemic uncertainty** comes from gaps in the model's knowledge, often because of limited or biased training data, and can be reduced by gathering more diverse data or improving the model.

The authors group existing methods into two broad families. **Probabilistic methods**—such as Bayesian neural networks, Monte Carlo dropout, Markov Chain Monte Carlo, bootstrap sampling, and deep ensembles—measure how confident a model is by looking at the spread or randomness in its predictions. These methods are good at spotting unusual cases and making predictions more reliable, but they are often too slow or computationally heavy for real-time clinical use.

To overcome these challenges, the review also looks at **non-probabilistic methods**, which aim to capture uncertainty without repeated model runs. Interval analysis can give upper and lower bounds for predictions, fuzzy logic can handle "in-between" cases at decision boundaries, and evidential approaches such as Dempster–Shafer theory can combine different types of evidence in a single forward pass. Another practical option is test-time augmentation, which checks how predictions change when the same image is slightly altered, giving a quick sense of reliability without major computational cost.

Because there are no standard datasets with "ground-truth" uncertainty labels, Huang et al. emphasize indirect ways of judging how well uncertainty methods work. These include checking whether predicted probabilities match real outcomes (calibration), measuring prediction spread (entropy, variance), using scoring rules such as the Brier score, and testing whether uncertain predictions align with cases that are misclassified or outside the model's training distribution. They also note how uncertainty can improve many tasks—such as segmentation, lesion detection, and semi-supervised training—by guiding models to focus on difficult or unreliable cases.

The review ends by pointing out several key gaps. There is still a lack of benchmark datasets that include uncertainty information, and no standard way to evaluate uncertainty across different methods. Practical tools for clinicians are also limited, making it difficult to bring these methods into routine workflows. Huang et al. call for closer links between uncertainty and explainability, so that doctors can see not just what the model predicts but also where and why it is uncertain. They also stress the importance of aligning these methods with clinical testing, ethical standards, and fairness goals.

These ideas directly influence the present thesis. Here, Monte Carlo Dropout and calibration are applied to CNN-based intracranial hemorrhage detection, producing slice-level uncertainty heatmaps and confidence estimates. This approach is designed to make the system's predictions more transparent, trustworthy, and useful for supporting decisions in emergency care.

6. **Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning**

Gal and Ghahramani [6] (2016) laid the theoretical groundwork for using dropout as a way to estimate uncertainty in deep learning models. They showed that applying dropout before every weight layer is mathematically similar to a Bayesian technique called variational inference in deep Gaussian processes. In this view, each dropout mask acts like a sample from an approximate distribution over the model's weights. This insight led to **Monte Carlo Dropout**, a simple but powerful method where a network trained with dropout can produce both predictions and estimates of its uncertainty by running multiple forward passes with dropout still active. This reframes dropout from being just a tool to prevent overfitting into a principled way of measuring uncertainty—an important step since standard neural networks often give overconfident predictions.

The authors built this connection by showing how dropout networks can be mapped to deep Gaussian processes through a spectral approximation of their covariance function. They proved that training a dropout network with L2 weight decay is essentially the same as performing approximate Bayesian inference, where the weight decay term represents the prior belief about the weights. This means that no special architecture or complex Bayesian machinery is needed—any existing dropout-trained network can provide uncertainty estimates just by keeping dropout active at test time.

Their experiments confirmed that Monte Carlo Dropout produces useful uncertainty estimates in practice. In regression tasks, such as predicting CO_2 levels from the Mauna Loa dataset, the method produced wider prediction intervals when extrapolating beyond the training data, similar to Gaussian Processes and unlike the overconfident predictions of standard models. In image classification on MNIST, MC Dropout flagged high uncertainty for rotated digits even when the softmax output gave misleadingly high confidence. On a range of UCI regression benchmarks, MC Dropout also outperformed more complex Bayesian neural network methods in both accuracy and uncertainty calibration, while remaining computationally efficient.

They also demonstrated the value of uncertainty in decision-making by applying MC Dropout to reinforcement learning. By using uncertainty estimates to guide exploration, their dropout-based Q-network achieved faster learning compared to the common ε-greedy strategy. This showed that uncertainty-aware models can not only improve prediction reliability but also make smarter decisions in interactive settings.

Overall, Gal and Ghahramani's work redefined dropout as more than just a regularizer. By grounding it in Bayesian theory, they showed that uncertainty estimates can be obtained from standard deep networks without changing their design or adding heavy computation. This breakthrough helped pave the way for safer and more trustworthy applications of deep learning, especially in fields where knowing when the model "does not know" is just as important as making accurate predictions.

7. **A Review of Uncertainty Estimation and its Application in Medical Imaging**

Zou [7](2023) take a close look at uncertainty estimation in medical imaging and argue that it is a key requirement for making AI systems safe to use in clinical care. They describe two main types of uncertainty: **aleatoric uncertainty**, which comes from unavoidable noise in the data such as low-quality scans or motion blur, and **epistemic uncertainty**, which reflects the model's lack of knowledge due to limited or biased training data. This distinction is especially relevant for detecting intracranial hemorrhage, where scan quality and patient anatomy can vary widely.

The authors outline a range of deep learning methods for handling uncertainty. These include **Bayesian neural networks**, **Monte Carlo Dropout**, **deep ensembles**, **test-time data augmentation**, and **conformal prediction**. Among these, Monte Carlo Dropout is highlighted as being particularly practical. By running multiple forward passes with dropout active, it can estimate epistemic uncertainty efficiently, making it suitable for real-time use. The review also emphasizes the role of **calibration methods** such as temperature scaling and isotonic regression, which make model probabilities more reliable and easier for clinicians to interpret.

Zou et al. stress the **clinical value** of uncertainty-aware models. By flagging unusual or out-of-distribution scans, these models can alert clinicians when predictions may be unreliable. They can also help triage borderline cases by passing uncertain results to human experts and provide visual tools such as **uncertainty heatmaps** that highlight regions of diagnostic doubt. Together, these approaches make model outputs more transparent and clinically useful.

At the same time, the review identifies several **challenges**. There is still no standard dataset or benchmark for evaluating uncertainty methods, which makes it hard to compare approaches. Some methods also require significant computation, slowing down clinical use. Most importantly, many studies stop at experimental demonstrations and do not fully address the realities of real-world deployment.

The insights from Zou et al. directly shape the present thesis. By adopting **Monte Carlo Dropout** and **calibration techniques**, this work aims to build a hemorrhage detection model that not only performs well but also provides **trustworthy confidence estimates**. In doing so, it seeks to improve both the safety and reliability of automated ICH detection in clinical practice.

8. **Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles**

Lakshminarayanan [8] (2017) present **deep ensembles** as a practical and scalable way to measure uncertainty in neural networks. Instead of relying on complex Bayesian methods, they train several independent models with different random starting points, data shuffles, or augmentations. At test time, the outputs from all models are combined, and the variation between them is used to estimate uncertainty.

The approach is easy to implement because it only requires training multiple models in parallel and averaging their predictions. This avoids the heavy computational cost of techniques like variational inference or Markov Chain Monte Carlo. In their experiments across standard benchmarks, including ImageNet, deep ensembles produced predictions that were not only accurate but also well-calibrated when judged with metrics like negative log-likelihood and the Brier score. Importantly, the method also showed higher uncertainty on out-of-distribution data, making it more reliable in situations where inputs differ from the training set—a key consideration for medical applications.

In this thesis, deep ensembles are explored as a **complementary strategy** to Monte Carlo Dropout. While MC Dropout estimates uncertainty by running multiple stochastic passes through a single model, deep ensembles capture variability across several independently trained models. By comparing the two methods in intracranial hemorrhage detection, this work aims to identify which provides the most trustworthy confidence estimates. Combining these approaches may strengthen the reliability of automated hemorrhage triage, helping ensure that uncertain cases are flagged for expert review and improving the overall safety of AI support in clinical practice.

9. **Uncertainty-aware deep-learning model for prediction of supratentorial hematoma expansion from admission non-contrast head computed tomography scan**

Tran [9](2024) developed an uncertainty-aware deep learning model to predict supratentorial hematoma expansion from admission non-contrast head CT scans, an important prognostic indicator for patient outcomes. Employing a DenseNet-121 backbone, the authors integrated **Monte Carlo Dropout** during inference to derive predictive uncertainty estimates. By applying **entropy-based thresholds**, the model stratified cases into high-confidence predictions and those warranting expert review. The approach achieved an AUC of 0.81 for forecasting ≥6 mL hematoma expansion and demonstrated that **high-confidence predictions** correlated significantly with adverse outcomes such as in-hospital mortality.

Despite these advances in uncertainty-guided prognostication, the study's focus on hematoma expansion limits its applicability to **initial intracranial hemorrhage detection**, and it does not leverage the standard **CQ500** dataset. Moreover, the absence of **slice-level uncertainty visualizations** restricts the model's utility for real-time diagnostic interpretation during acute triage.

The current thesis addresses these gaps by designing a deep learning framework for **slice-level ICH detection** on CQ500 scans, incorporating Monte Carlo Dropout to quantify predictive uncertainty. A primary contribution is the generation of **slice-specific uncertainty heatmaps**, enabling clinicians to pinpoint ambiguous regions and make informed decisions about cases requiring specialist review. By extending Tran et al.'s uncertainty estimation paradigm to the **initial detection** task, this work seeks to enhance both diagnostic accuracy and clinical trust in automated hemorrhage triage systems.

10. Bayesian Deep Learning Outperforms Clinical Trial Estimators of Intracerebral and Intraventricular Hemorrhage Volume

Sharrock [10](2022) used Bayesian deep learning with Monte Carlo Dropout to segment intracerebral hemorrhage (ICH) and intraventricular hemorrhage (IVH) on CT scans from the MISTIE and CLEAR clinical trials. Their model produced accurate volumetric segmentations, with Dice similarity scores of 0.946 for ICH and 0.863 for IVH, and very strong correlations with ground truth volumes (0.994 and 0.980, respectively). A key finding was that the model's estimated uncertainty was closely tied to segmentation quality: higher uncertainty strongly correlated with lower segmentation accuracy ($r = -0.849$) and greater volume error ($r = 0.735$). This shows that uncertainty measures can reliably signal when the model's predictions may be less trustworthy.

However, the study was focused only on volumetric segmentation and did not tackle the earlier stage of detecting hemorrhage on individual CT slices. It also did not make use of the CQ500 dataset, nor did it provide slice-level uncertainty visualizations that would be more useful in real-time diagnostic settings such as emergency departments.

This thesis builds on Sharrock et al.'s work by applying Monte Carlo Dropout to slice-level hemorrhage detection in the CQ500 dataset. Instead of focusing only on volumetric outcomes, it generates slice-specific uncertainty heatmaps that highlight ambiguous regions at the point of initial detection. This approach aims to support clinicians directly during interpretation, improving diagnostic confidence and trust in automated systems from the earliest stage of hemorrhage assessment.

10. Integrating uncertainty in deep neural networks for MRI based stroke analysis

Herzog [11](2020) developed a Bayesian convolutional neural network to detect ischemic stroke lesions on MRI scans, using Monte Carlo Dropout during both training and inference to estimate uncertainty. In a study of 511 patients, adding uncertainty estimates improved slice-level diagnostic accuracy to 95.33% and raised patient-level reliability to 95.89%. Importantly, cases with incorrect predictions showed higher uncertainty scores, which meant the model could flag ambiguous cases for expert review and improve overall diagnostic safety.

While this work is influential for MRI-based stroke analysis, its focus is quite different from the present thesis. Herzog et al. concentrated on volumetric segmentation and patient-level decision support in MRI, rather than slice-level detection on CT scans. Their framework did not generate slice-specific uncertainty heatmaps, nor did it use hemorrhage-specific datasets like CQ500, which limits its relevance for intracranial hemorrhage triage in emergency CT workflows.

The current thesis builds on Herzog et al.'s uncertainty-aware approach but adapts it to CT-based hemorrhage detection. By applying Monte Carlo Dropout to individual CT slices from the CQ500 dataset, this work produces slice-level uncertainty heatmaps that highlight ambiguous regions right at the point of detection. In doing so, it translates the strengths of Bayesian uncertainty into a CT setting, bridging the gap between MRI segmentation studies and CT classification tasks. This approach aims to improve interpretability, safety, and clinical trust during the earliest stage of hemorrhage assessment.

11. Enhancing Monte Carlo Dropout Performance for Uncertainty Quantification

Asgharnezhad [12](2025) introduced an improved version of Monte Carlo Dropout aimed at making uncertainty estimates in deep learning models more reliable. Their key innovation was an uncertainty-aware loss function that discourages the model from being overconfident when wrong and rewards it for being confident when correct. This design helped align the model's predictive entropy with its actual accuracy, producing uncertainty scores that better reflected reality. To further improve performance, the authors used meta-heuristic optimization methods—including the Grey Wolf Optimizer, Bayesian Optimization, and Particle Swarm Optimization—to automatically fine-tune hyperparameters such as dropout rate and entropy thresholds. Together, these strategies reduced calibration errors and improved a new metric they called "uncertainty accuracy," which measures how well the model's confidence matches its true correctness.

Despite these advances, the study did not focus on medical imaging or intracranial hemorrhage detection. Instead, the experiments were carried out on general-purpose datasets outside the clinical domain, without applications to CT scan analysis or uncertainty heatmaps. This means that, while the work strengthened the theoretical and practical foundations of Monte Carlo Dropout, it did not directly test its value in healthcare contexts.

The present thesis adapts Monte Carlo Dropout specifically for intracranial hemorrhage detection using the CQ500 dataset. Unlike Asgharnezhad et al., the focus here is on slice-level CT analysis, with the added contribution of generating uncertainty heatmaps that highlight ambiguous regions for clinical review. Although advanced hyperparameter tuning techniques were not incorporated, the project keeps the same central aim: making model confidence more trustworthy and aligned with actual performance. In the future, extending this work with adaptive dropout tuning could further refine uncertainty calibration for hemorrhage detection in CT imaging.

12. Deep Learning Applied to Intracranial Hemorrhage Detection

Cortés-Ferre [13] (2023) proposed a deep learning system for detecting intracranial hemorrhage at the slice level by combining a ResNet backbone with the lightweight EfficientDet-D0 object detector and using Grad-CAM for visual explanations. Tested on their proprietary dataset, the model achieved 92.7% accuracy and an AUC of 0.978. Importantly, the framework was designed with "Green AI" principles in mind, ensuring efficiency and making it suitable for real-time clinical use. Grad-CAM heatmaps added interpretability by highlighting the image regions that influenced the model's predictions, helping clinicians quickly focus on the most relevant areas during triage.

However, while the model improved interpretability, it did not include uncertainty quantification. In other words, it could not measure how confident it was in its predictions or flag ambiguous cases that may require expert review. In addition, the system was not evaluated on the widely used CQ500 dataset, so its ability to generalize across different institutions and CT scanners remains unknown.

The current thesis takes these gaps into account by integrating Monte Carlo Dropout into a CNN-based ICH detection model trained on CQ500. This approach goes beyond visual explanations by producing both calibrated confidence estimates and slice-level uncertainty heatmaps that explicitly mark regions of diagnostic doubt. By combining interpretability with reliable uncertainty measures, the proposed method not only highlights suspicious areas but also communicates how much trust can be placed in each prediction. This helps guide clinical oversight more effectively and supports safer decision-making in emergency CT triage.

13. Intracranial hemorrhage segmentation and classification framework in computer tomography images using deep learning techniques

Ahmed and Prakasam [14] (2025) introduced **IHSNet**, a deep learning framework designed for intracranial hemorrhage (ICH) segmentation and subtype classification on CT scans. Their approach combines a MUNet (multiclass U-Net) for identifying and segmenting hemorrhage regions with fully connected layers for classifying five ICH subtypes: intraventricular hemorrhage (IVH), epidural hemorrhage (EDH), intraparenchymal hemorrhage (IPH), subdural hemorrhage (SDH), and subarachnoid hemorrhage (SAH). On a private dataset, IHSNet reported strong results, achieving 98.53% segmentation accuracy and 98.71% classification accuracy, with Dice coefficients between 0.64 and 0.92 across different subtypes. These outcomes highlight its potential as a high-performing automated ICH diagnosis system.

Despite these strengths, IHSNet does not incorporate uncertainty quantification. The model offers no way of communicating how confident it is in its predictions or of flagging ambiguous regions for further expert review—an important safeguard in high-risk medical applications. In addition, IHSNet has not been validated on the widely used CQ500 dataset, leaving open questions about its robustness across different scanners and institutions. The lack of slice-level uncertainty visualizations further reduces its interpretability and limits its immediate applicability to real clinical workflows where transparency and reliability are essential.

The present thesis seeks to address these gaps by applying Monte Carlo Dropout to a CNN-based detection model trained on the CQ500 dataset. Rather than focusing only on whether a hemorrhage is present, this approach produces slice-level uncertainty heatmaps that highlight areas where the model is less confident. These visualizations enhance traditional segmentation and classification by pairing predictions with explicit confidence information, making it easier for clinicians to judge when expert review is necessary. By extending IHSNet's strong performance with calibrated uncertainty estimation, this thesis aims to improve diagnostic safety and provide more trustworthy decision support in emergency ICH triage.

14. Intracerebral hemorrhage segmentation in emergency settings

Choi and colleagues [15] addressed the urgent need for rapid and reliable intracerebral hemorrhage (ICH) segmentation in emergency settings by developing an uncertainty-aware deep learning framework. They trained their model on a diverse dataset of 1,200 non-contrast head CT scans from three hospitals, accounting for variations in scanners, slice thickness, and acquisition artifacts. By incorporating residual connections into a U-Net architecture, they improved gradient stability and segmentation accuracy for both large and subtle bleeds.

A key innovation of their work was the integration of uncertainty estimation. Using Monte Carlo Dropout at test time, the model generated multiple outputs per slice to produce voxel-level confidence maps, highlighting ambiguous regions such as low-contrast or partial-volume areas. Test-time augmentation further strengthened these estimates. Clinically, this allowed low-confidence slices to be flagged for expert review, raising effective Dice performance from 0.87 to 0.91. Importantly, the framework operated in real time, processing each slice in under 200 milliseconds, which meets emergency triage requirements.

Choi et al.'s study exemplifies a human–AI collaboration model, where confident predictions support rapid decision-making and uncertain cases are deferred to radiologists. Building on this, the present thesis extends the principle of uncertainty-aware modeling from volumetric segmentation to slice-level detection on the CQ500 dataset. It generates slice-specific uncertainty maps, evaluates calibration of confidence scores, and maintains real-time inference, aiming to create deployable triage tools that balance speed with clinical reliability.

15. Semantic Segmentation of Spontaneous Intracerebral Hemorrhage

Hu [16] (2022) conducted a large-scale evaluation of deep learning–based segmentation for intracerebral hemorrhage (ICH), perihematomal edema (PHE), and intraventricular hemorrhage (IVH) using a multicenter dataset from the TICH-2 trial (n = 1,732). Employing a three-dimensional nnU-Net architecture, the study systematically compared loss functions to address class imbalance, particularly for IVH, which appeared in only 30% of cases. Their best-performing models achieved strong Dice scores—0.92 for ICH, 0.66 for PHE, and up to 1.00 for IVH, and consistently outperformed BLAST-CT and DeepLabv3+. The use of Focal loss proved especially effective for IVH, delivering superior segmentation consistency and volumetric agreement. These results reaffirm the robustness of U-Net–based methods across heterogeneous imaging protocols and highlight tailored loss design as a practical strategy for handling rare lesion types.

However, while excelling in segmentation accuracy, the framework lacks uncertainty estimation, real-time deployment analysis, and broader validation beyond a single train–test split. For clinical translation, uncertainty-aware extensions—for example, Monte Carlo Dropout with slice-level heatmaps—could complement the accuracy of U-Net by flagging ambiguous boundaries in PHE and IVH, thereby improving safety and interpretability in acute ICH triage.

16. AI-Powered Intracranial Hemorrhage Detection: A Co-Scale Convolutional Attention Model with Uncertainty-Based Fuzzy Integral Operator and Feature Screening

Hosseini Chagahi [17](2024) introduced a two-stage uncertainty-aware framework for detecting intracranial hemorrhage (ICH) in CT scans. In the first stage, they extended the Co-Scale Convolutional Attention (CCA) model to extract features from all slices. Principal component analysis (PCA) was then used to reduce the features to the most informative ones, which were further refined using a bootstrap forest and a boosting neural network. This process created a compact and explainable feature space that emphasized the most discriminative slice-level patterns.

In the second stage, the authors proposed an uncertainty-based fuzzy integral method to combine slice-level predictions into a scan-level decision. This fusion step considered both the predictive confidence of each slice, estimated through entropy, and dependencies across slices. By weighting slices according to both certainty and spatial continuity, the model achieved more reliable scan-level predictions than simpler methods like averaging or max-pooling.

Evaluated on a proprietary dataset of about 4,000 scans, the model achieved 96.2% accuracy and a 0.983 ROC-AUC, outperforming both the baseline CCA model and Monte Carlo Dropout ensembles. The feature screening step reduced dimensionality by around 80% with little loss of accuracy, while the fuzzy fusion further improved performance. The study also showed that attention maps and confidence maps aligned well with radiologist annotations, highlighting regions of diagnostic uncertainty.

Despite these strengths, the framework was only tested on a single dataset and not on public benchmarks such as CQ500, which limits evidence of generalizability. Moreover, uncertainty was only applied during fusion and not in earlier steps such as feature selection or slice referral, and the fuzzy operator added hyperparameters that could complicate real-time use.

For the present thesis, this study is relevant because it demonstrates how uncertainty can be used not only for prediction but also for decision fusion. Combining such fuzzy confidence weighting with slice-level uncertainty maps from Monte Carlo Dropout could provide a stronger and more flexible framework for ICH detection on datasets like CQ500.

17. Development and External Validation of a Deep Learning Algorithm to Identify Subarachnoid Hemorrhage on CT Scans

Albers [18](2023) developed and externally validated a 3D ResNet-based model for rapid detection of subarachnoid hemorrhage (SAH) on noncontrast head CT scans. Trained on 8,500 annotated scans from four centers, the model was tested on two independent cohorts and achieved strong performance, with a pooled ROC-AUC of 0.961, sensitivity of 92.4%, and specificity of 88.7%. Saliency maps showed that the network localized typical SAH regions, while error analysis revealed that very small bleeds were often missed and artifacts sometimes caused false positives.

Despite its accuracy, the framework has several limitations: it focuses solely on SAH, excludes other hemorrhage types, and lacks uncertainty quantification, offering only binary outputs without confidence scores. This restricts its ability to flag ambiguous or high-risk cases for radiologist review. For the present thesis, Albers et al.'s work highlights the value of large, multi-center validation, but also motivates extending beyond single-class detection. Incorporating Monte Carlo Dropout into such architectures on the CQ500 dataset could provide both multi-label ICH detection and slice-level uncertainty maps, enhancing clinical trust and supporting safer, uncertainty-aware deployment in emergency triage.

18. Deep learning-assisted detection and segmentation of intracranial haemorrhage in noncontract computed tomography scans of acute stroke patients: a systematic review and meta-analysis

Intracranial haemorrhage [19](ICH) is a critical neurological emergency where timely diagnosis directly impacts patient outcomes. Non-contrast computed tomography (NCCT) remains the gold standard for detection, yet manual interpretation can be time-consuming and subject to inter-rater variability. Recent advances in artificial intelligence, particularly deep learning (DL), have shown promise in automating ICH detection and segmentation with accuracy comparable to human experts, while offering the advantage of rapid processing.

A recent systematic review and meta-analysis synthesized evidence from 36 original studies evaluating DL algorithms for ICH detection and segmentation. The pooled results demonstrated strong performance, with sensitivity of 0.89, specificity of 0.91, and an area under the receiver operating characteristic curve (AUROC) of 0.94. Additionally, DL approaches achieved a Dice Similarity Coefficient (DSC) of 0.84 for segmentation tasks, indicating robust agreement with manual annotations. Notably, DL models performed comparably to radiologists in quantification accuracy, while providing significant improvements in efficiency.

These findings underscore the potential of DL-assisted tools to enhance diagnostic workflows in emergency neuroimaging. However, the review also highlighted limitations, including a lack of multicentre randomized controlled trials to establish generalizability and clinical safety. As such, while DL methods are emerging as valuable adjuncts in ICH detection and segmentation, further validation is essential before their widespread adoption in routine clinical practice.

CHAPTER 3- METHODOLOGY

3.1 Data Collection

3.1.1 Dataset Characteristics

The experiments in this study are based on the CQ500 dataset which was downloaded from academic torrents [20], a publicly accessible collection of head CT scans annotated by expert radiologists for the presence of intracranial haemorrhage, as shown in **Figure 1 CQ500 dataset.** After preprocessing and quality filtering, the final dataset consisted of 489 patients with complete imaging and corresponding labels. In total, these scans yielded 168,590 axial slices, averaging approximately 345 slices per patient. One patient was excluded because of missing or incomplete imaging data.

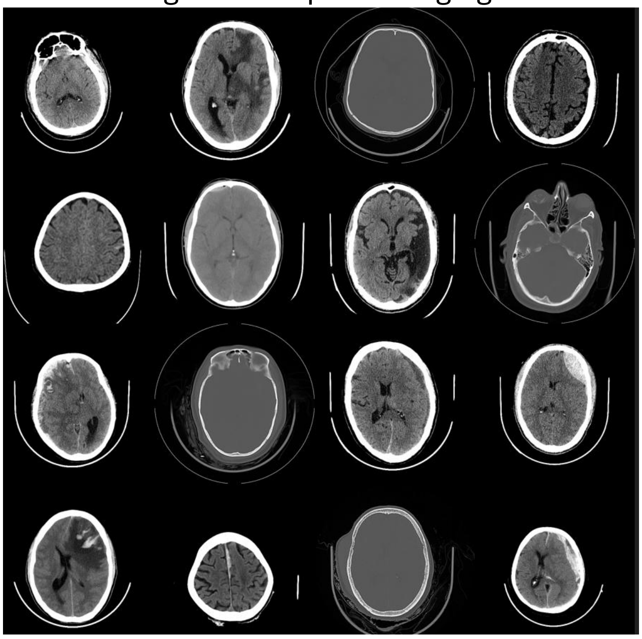

Figure 1 CQ500 dataset

Figure 2 (Radiologist Haemorrhage Analysis) illustrates the variability among the three radiologists who provided labels. Radiologist 1 identified haemorrhage in 230 patients (46.8%), Radiologist 2 in 196 patients (39.9%), and Radiologist 3 in 193 patients (39.3%). Such variation highlights the inherent subjectivity of radiological assessment, particularly in cases where haemorrhage is subtle or ambiguous.

To harmonize the labels, two consensus strategies were employed. In the *Label_AtLeastOne* approach, a case was classified as positive if any of the three radiologists reported haemorrhage. This produced 244 positive patients (49.9%) and 245 negatives (50.1%). In the stricter *Label_AtLeastTwo* approach, at least two radiologists had to agree for a case to be marked positive, resulting in 205 haemorrhage cases (41.8%) and 284 non-haemorrhage cases (58.2%).

Overall, full agreement among all three annotators was achieved in 413 out of 489 patients (84.1%). The remaining 76 cases (15.9%) showed partial disagreement, with 41 patients (8.4%) marked positive by only one radiologist and 37 patients (7.5%) marked positive by two. These discordant cases likely represent the most diagnostically challenging instances in the dataset.

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============================================================
MULTI-RADIOLOGIST HEMORRHAGE ANALYSIS
============================================================
Analyzing 3 radiologists: ['ICH_R1', 'ICH_R2', 'ICH_R3']
Total patients: 491

INDIVIDUAL RADIOLOGIST ASSESSMENTS:
----------------------------------------
ICH_R1         : 230 hemorrhage cases ( 46.8%)
ICH_R2         : 196 hemorrhage cases ( 39.9%)
ICH_R3         : 193 hemorrhage cases ( 39.3%)

CONSENSUS (MAJORITY VOTE):
----------------------------------------
Hemorrhage cases: 205 (41.8%)
No hemorrhage cases: 286 (58.2%)

AGREEMENT STATISTICS:
----------------------------------------
Full agreement among all radiologists: 413 patients (84.1%)

AGREEMENT DISTRIBUTION (how many radiologists said hemorrhage):
  0 radiologists: 245 patients ( 49.9%)
  1 radiologists:  41 patients (  8.4%)
  2 radiologists:  37 patients (  7.5%)
  3 radiologists: 168 patients ( 34.2%)
```

Figure 2 Radiologist Haemorrhage Analysis

The large number of slices per subject is a direct consequence of CT's volumetric acquisition, where multiple axial views are required to cover the entire brain. This design enables analysis not only at the patient level but also at the slice level, giving flexibility in how models are developed and evaluated. The substantial slice count per patient also ensures that deep learning algorithms can be trained on a rich supply of data.

With 489 patients, the dataset is suitably sized for this investigation. It provides sufficient cases to support reliable cross-validation yet still mirrors the practical limitations of medical imaging studies, which often rely on relatively modest sample sizes. Notably, under the *Label_AtLeastOne* consensus strategy, the dataset achieves a nearly even split between positive (49.9%) and negative (50.1%) cases. This balance reduces the need for extensive rebalancing procedures and creates favourable conditions for machine learning model training.

3.1.2 DICOM Series Loading and Anatomical Ordering

As illustrated in **Figure 3 (Raw Dataset Before Processing)**, the original DICOM files were organized into patient-specific directories and systematically converted to obtain both image pixel data and accompanying metadata. Each scan series was checked for completeness and consistency, with automated safeguards in place to manage corrupted images, missing attributes, or format discrepancies issues that frequently arise in clinical imaging archives.

Within each patient's volumetric study, the slices were arranged according to their *Instance Number* metadata field, ensuring the correct anatomical order. Preserving this sequence was essential for maintaining spatial continuity between adjacent slices and for representing the brain volume accurately throughout the dataset. (**Figure 4 - After Processing)**

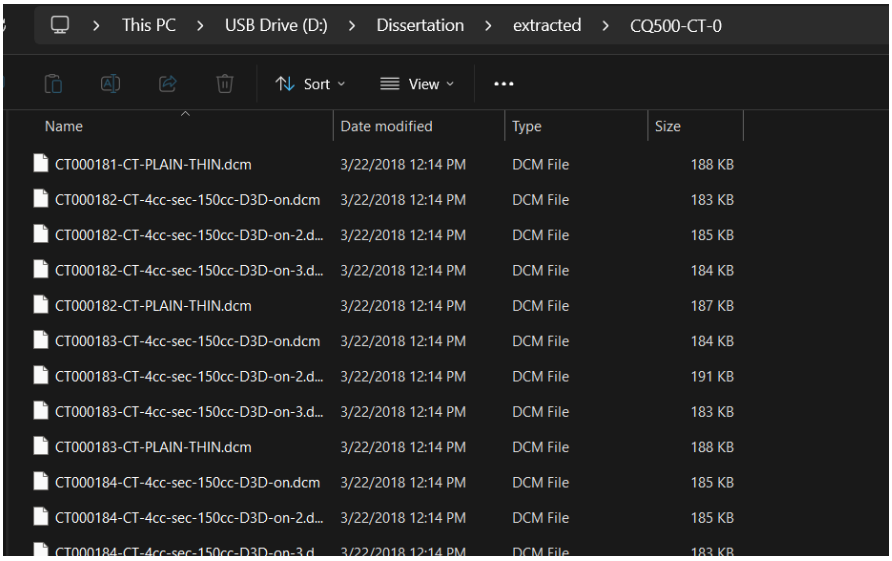

Figure 3 Raw Dataset Before Processing

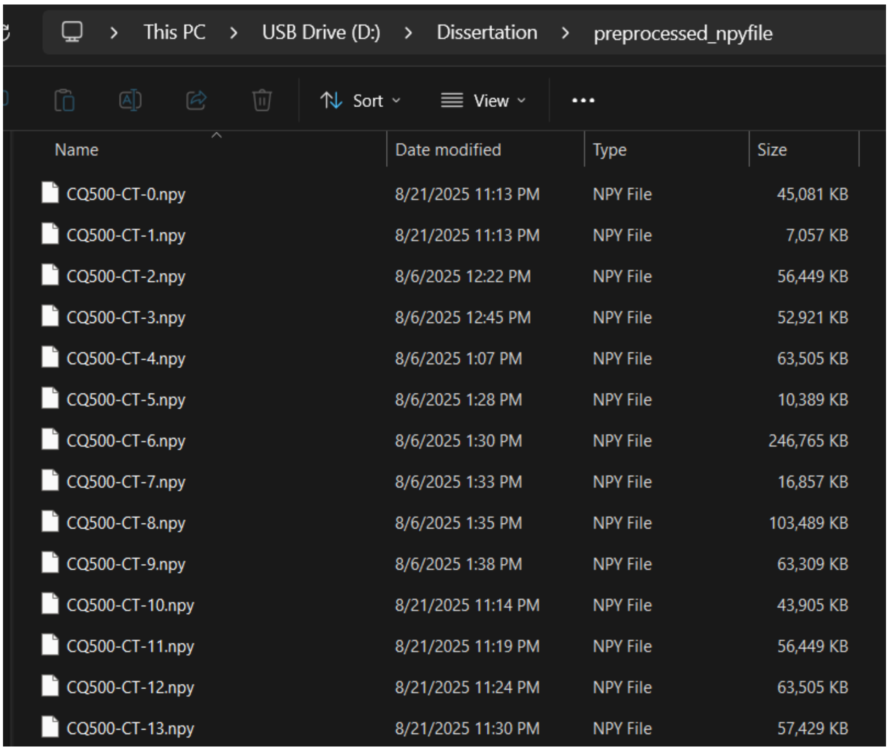

Figure 4 After Processing

3.1.3 Intensity Normalization and Standardization

Medical imaging datasets often show large variations in pixel intensity caused by differences in scanner hardware, acquisition settings, and protocol design. To reduce this variability, a slice-level normalization procedure was applied. For each axial slice, pixel values were independently rescaled using min–max normalization, defined as:

$$I_{normalizes}(x,y) = \frac{I_{original}(x,y) - I_{min}}{I_{max} - I_{min}}$$

where $I_{original}(x,y)$ is the raw pixel intensity at location (x,y), and I_{min} and I_{max} represent the minimum and maximum intensities within that slice.

This slice-based strategy was selected instead of whole-volume normalization to better preserve contrast variations that can highlight subtle haemorrhages. In this way, the normalization maintained intra-slice intensity relationships while ensuring that all slices shared a consistent dynamic range across the dataset. [21]

3.1.4 Spatial Standardization and Resolution Harmonization

To standardize input dimensions across the dataset, all CT images were resized to 224 × 224 pixels using area-based interpolation, as shown in **Figure 5 - CT slice before and after resizing**. This resolution was selected because it is widely compatible with popular convolutional neural network architectures, balances computational efficiency, and preserves sufficient anatomical detail for haemorrhage detection. [22]

Area interpolation was specifically employed during downsampling as it minimizes aliasing artifacts and maintains edge clarity, both of which are important for highlighting haemorrhages that typically manifest as sharp, high-contrast regions on CT scans. Additionally, resizing addressed inconsistencies in matrix size, pixel spacing, and field of view that naturally arise from variations in scanners and acquisition protocols, thereby ensuring uniformity across the dataset.

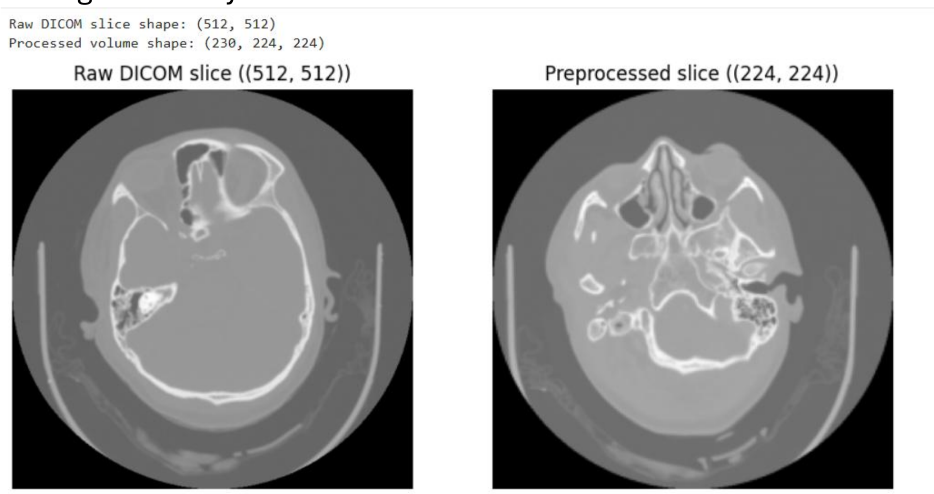

Figure 5 - CT Slice before and after resizing

3.1.5 Ground Truth Label Processing and Inter-Rater

The dataset included haemorrhage assessments from three radiologists, which required processing to handle differences in their interpretations. Each case was originally labelled as either having or not having haemorrhage. [23]

Two labelling methods were created, as shown in **Figure 6 - Consensus Label processed dataset**, which had:

- **Consensus-OR (Label_AtLeastOne):** A case was marked positive if at least one radiologist reported haemorrhage. This increases sensitivity by capturing even subtle or uncertain cases.
- **Consensus-Majority (Label_AtLeastTwo):** A case was marked positive only if at least two radiologists agreed. This increases specificity by focusing on cases with stronger agreement.

Using both methods helps manage uncertainty in medical imaging. The Consensus-OR approach suits screening situations where missing a haemorrhage would be risky, while the Consensus-Majority approach fits diagnostic settings where avoiding false alarms is important. Comparing the two provides insights into how diagnostic uncertainty affects decision-making.

	A	B	C	D	E	F	G
1	PatientID	ICH_R1	ICH_R2	ICH_R3	Label_AtLeastOne	Label_AtLeastTwo	
2	CQ500-CT-427	1	1	1	1	1	
3	CQ500-CT-181	1	1	1	1	1	
4	CQ500-CT-99	0	0	0	0	0	
5	CQ500-CT-47	0	0	0	0	0	
6	CQ500-CT-195	0	0	0	0	0	
7	CQ500-CT-357	0	0	0	0	0	
8	CQ500-CT-80	1	1	1	1	1	
9	CQ500-CT-2	0	0	0	0	0	
10	CQ500-CT-361	0	0	0	0	0	
11	CQ500-CT-412	1	0	1	1	1	
12	CQ500-CT-401	0	0	0	0	0	
13	CQ500-CT-342	1	1	1	1	1	
14	CQ500-CT-469	1	1	1	1	1	
15	CQ500-CT-57	1	1	1	1	1	
16	CQ500-CT-155	1	1	1	1	1	
17	CQ500-CT-273	1	1	1	1	1	
18	CQ500-CT-423	1	1	0	1	1	
19	CQ500-CT-282	0	0	0	0	0	
20	CQ500-CT-149	1	1	1	1	1	
21	CQ500-CT-211	1	1	1	1	1	
22	CQ500-CT-420	1	1	1	1	1	
23	CQ500-CT-196	1	1	1	1	1	
24	CQ500-CT-488	1	1	1	1	1	
25	CQ500-CT-277	1	0	1	1	1	

Figure 6 - Consensus Label processed dataset

3.1.6 Contrast Enhancement Using Adaptive Histogram Equalization

Contrast Limited Adaptive Histogram Equalization (CLAHE) was used to improve local contrast while avoiding the noise problems of global histogram equalization. Intracranial hemorrhages can appear as subtle density differences on CT scans, especially in small or isodense bleeds. CLAHE helps highlight these subtle changes by adjusting contrast locally while keeping normal brain tissue clear. Its adaptive design ensures consistent enhancement across different brain regions with varying tissue density, as shown in **Figure 7 - Processed and Clahe enhanced Slice comparision** The image is divided into small tiles, and histogram equalization is applied within each tile, with a contrast limit to prevent over-enhancement.The clip limit C*C* and tile grid size T*T* were set empirically to 2.0 and 8×8, respectively. [24]

$$CLAHE_{tile}(p) = \min\,(Hist(p), C) / \sum_{p} \min\,(Hist(p), C)$$

These values were selected after testing on neuroimaging data to balance clearer contrast with minimal artifacts.

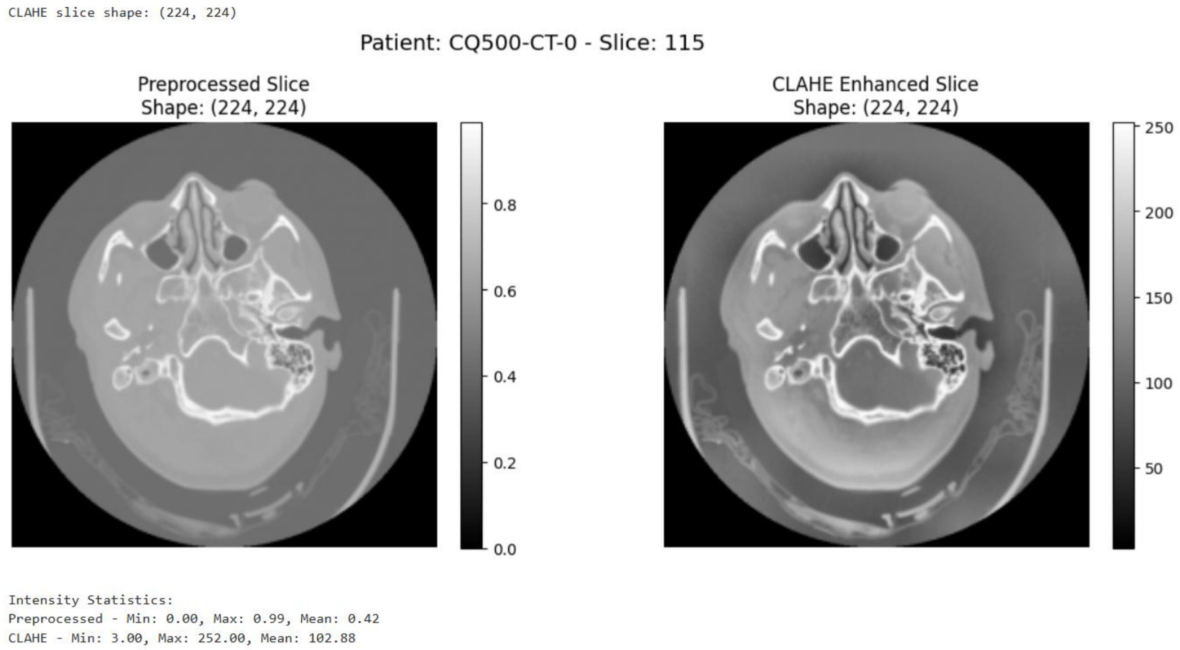

Figure 7 - Processed and Clahe enhanced Slice comparison

3.1.7 Traditional Feature Extraction for Baseline Comparison

A multi-dimensional feature extraction pipeline was implemented to provide baseline benchmarks and to facilitate comparison with deep learning methods. The approach incorporated several categories of descriptors, each designed to capture distinct aspects of CT image characteristics relevant for haemorrhage identification.

First-order intensity features were computed to summarize overall pixel distributions within each slice. The mean value represented the average brightness level, while the standard deviation quantified variability in intensities. These measures are useful since haemorrhagic regions typically appear brighter than normal parenchyma.

Histogram-based descriptors were also generated to capture broader distribution patterns. Ten equally spaced bins were used across the normalized intensity range, and each histogram was normalized, allowing comparison across scans with different brightness levels. These features emphasize recurring intensity profiles that may indicate haemorrhagic tissue.

Structural features were added to characterize boundary and region properties. Edge density was measured using Canny edge detection with thresholds of 100 and 200, reflecting the amount of sharp structural detail present. In addition, Otsu's method was applied to segment brighter regions, yielding an area estimate potentially corresponding to haemorrhagic lesions.

The Otsu threshold T_{otsu} was computed as:

$$T_{otsu} = \arg max_t \sigma_B^2(t)$$

where $\sigma_B^2(t)$epresents the between-class variance for threshold *t*.

The complete feature vector for each image slice was formulated as:

$$f = [\mu, \sigma, \ E_{count} \ , A_{otsu}, h_1, h_2, \dots \dots . h_{10}]^T$$

where each component represents a specific image characteristic as described above, resulting in a 14-dimensional feature space.Patient-level features were then generated by averaging across slices (mean pooling), ensuring robustness to slice variations while retaining overall diagnostic characteristics.

Finally, quality control was applied by removing slices with very low mean intensity (<0.01 on the normalized scale), as these usually represent empty regions or artifacts. This preprocessing pipeline established consistent, reliable features suitable for further analysis, as shown in **Figure 8 - Dataset with the extracted Features**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	PatientID	Label_AtLe	Label_AtLe	mean_inte	std_intensi	edge_coun	otsu_area	hist_bin_0	hist_bin_1	hist_bin_2	hist_bin_3	hist_bin_4	hist_bin_5	hist_bin_6	hist_bin_7	hist_bin_8	hist_bin_9
2	CQ500-CT-0	1	0	0.394197	0.231713	4155.257	39222.12	0.210458	0.003042	0.017597	0.196655	0.248543	0.139662	0.116312	0.042043	0.016536	0.009153
3	CQ500-CT-1	1	1	0.404578	0.232585	2855.5	39020.75	0.217865	0.002606	0.009355	0.108112	0.337908	0.111722	0.163149	0.024728	0.016058	0.008498
4	CQ500-CT-10	1	1	0.400807	0.233175	3526.554	39365.92	0.210035	0.002932	0.012489	0.163203	0.275989	0.138463	0.140654	0.023597	0.018591	0.014045
5	CQ500-CT-100	0	0	0.193476	0.218884	3053.266	17125.14	0.605186	0.035767	0.029497	0.122155	0.12309	0.028133	0.014714	0.013665	0.016287	0.011506
6	CQ500-CT-101	0	0	0.383799	0.225419	3646.89	39392	0.20955	0.002823	0.025825	0.232551	0.2235	0.14456	0.104397	0.029154	0.018429	0.009212
7	CQ500-CT-102	1	1	0.403294	0.240308	3430.074	38712.57	0.220135	0.002911	0.016165	0.147908	0.252562	0.145099	0.148187	0.029561	0.021264	0.016208
8	CQ500-CT-103	0	0	0.425627	0.245474	2974.569	39392.55	0.209165	0.002549	0.007383	0.110744	0.277247	0.11251	0.212259	0.025858	0.01896	0.023326
9	CQ500-CT-104	0	0	0.371408	0.224277	3655.42	39126.48	0.213576	0.003318	0.038624	0.266004	0.185463	0.16168	0.076321	0.021951	0.020288	0.012776
10	CQ500-CT-105	0	0	0.420627	0.242912	2975.434	39257.51	0.209538	0.00274	0.009296	0.1185	0.253345	0.153936	0.184568	0.021583	0.027296	0.019197
11	CQ500-CT-106	1	1	0.413721	0.238335	2978.98	39318.69	0.209394	0.002626	0.011044	0.150332	0.250812	0.142736	0.163275	0.028381	0.023329	0.018071
12	CQ500-CT-107	1	1	0.41494	0.238337	3371.5	39413.22	0.209363	0.002462	0.007098	0.140565	0.270399	0.114633	0.193565	0.030757	0.019604	0.011556
13	CQ500-CT-108	1	0	0.39646	0.229827	3572.83	39404.2	0.209709	0.00249	0.011994	0.176259	0.269733	0.140309	0.137058	0.024323	0.016972	0.011153
14	CQ500-CT-109	1	1	0.405574	0.234867	3448.194	39309.9	0.209634	0.002641	0.012451	0.160155	0.260489	0.134317	0.158252	0.030792	0.019459	0.011811
15	CQ500-CT-11	1	0	0.432336	0.252336	2872.292	38775.18	0.222377	0.002574	0.004396	0.062269	0.24633	0.191292	0.1588	0.068137	0.029842	0.013982
16	CQ500-CT-110	0	0	0.407877	0.23266	3160.213	39422.79	0.209472	0.002535	0.00545	0.122329	0.316373	0.145252	0.142398	0.028168	0.015126	0.012898
17	CQ500-CT-111	1	1	0.407046	0.239415	3734.602	38996.8	0.209815	0.00333	0.023984	0.175671	0.198803	0.182808	0.128158	0.032383	0.026701	0.018345
18	CQ500-CT-112	0	0	0.187915	0.210037	3269.663	16777.97	0.601263	0.047094	0.036566	0.112225	0.122782	0.032074	0.01551	0.013481	0.012829	0.006175
19	CQ500-CT-113	1	1	0.354809	0.22843	3407.228	37297.99	0.251587	0.002832	0.031787	0.253467	0.185776	0.140105	0.088039	0.024544	0.0151	0.006764
20	CQ500-CT-114	1	1	0.173719	0.218485	2987.082	13427.57	0.6549	0.054543	0.029924	0.049274	0.125934	0.033246	0.015562	0.010781	0.013788	0.012048
21	CQ500-CT-115	1	1	0.159242	0.197344	3040	13149.04	0.666895	0.053827	0.028181	0.085514	0.101559	0.023399	0.012663	0.011341	0.010823	0.005799
22	CQ500-CT-116	0	0	0.358018	0.237182	3506.086	36531.56	0.26431	0.00342	0.021355	0.203693	0.229971	0.140759	0.085177	0.023786	0.016889	0.010641
23	CQ500-CT-117	0	0	0.401786	0.227574	3066.245	39443.45	0.209326	0.002438	0.005502	0.11506	0.345585	0.141622	0.12712	0.026694	0.018095	0.008558
24	CQ500-CT-118	1	1	0.183132	0.211016	3382.776	14991.16	0.610513	0.066237	0.040966	0.07996	0.122751	0.025773	0.016948	0.014942	0.014618	0.007293

Figure 8 Dataset with extracted Features

3.2 Data Understanding and Analysis

3.2.1 Label Distribution

Figure 9 illustrates the distribution of hemorrhage types within the dataset, which includes labels for multiple categories of intracranial bleeding and related conditions. Although hemorrhage cases are well represented overall, the frequency of individual subtypes varies considerably. Some appear often, while others are rare but clinically significant. This imbalance complicates model training, as algorithms tend to favor the majority classes and may struggle to recognize less common findings. To mitigate this issue and promote more balanced performance, strategies such as applying class-weighted loss functions, using stratified sampling, and introducing targeted data augmentation were explored.

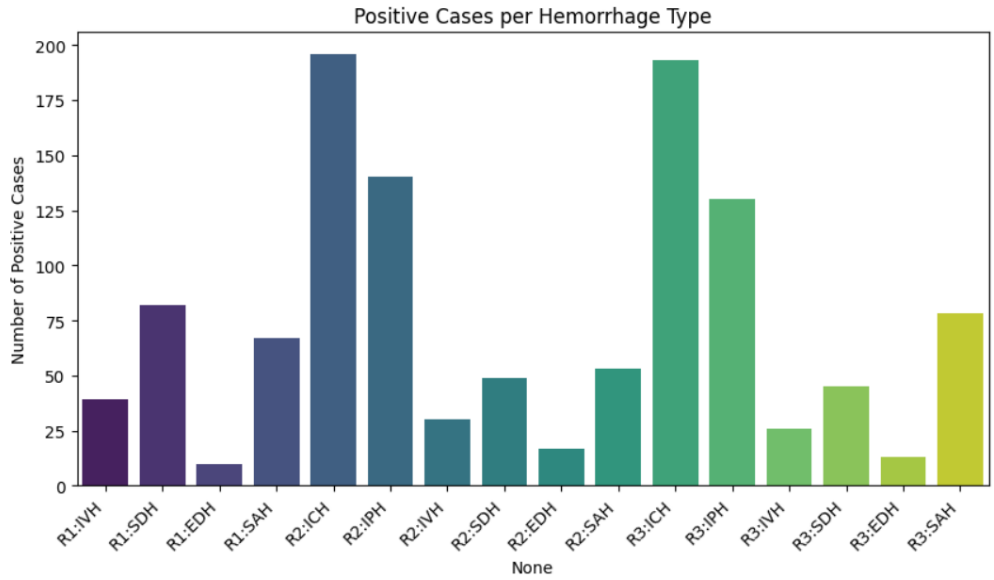

Figure 9 Hemorrage Distribution

3.2.2 Correlation Between Haemorrhage Subtypes

Analysis of the dataset reveals that certain haemorrhage subtypes frequently co-occur, whereas others are typically observed in isolation (Figure 2: Correlation Matrix). These associations align with clinical understanding, where overlapping vascular mechanisms or anatomical proximity can give rise to multiple haemorrhage patterns within the same patient. Identifying such relationships is valuable for model development and evaluation, as it encourages algorithms to recognize both shared features and unique characteristics of different haemorrhage presentations.

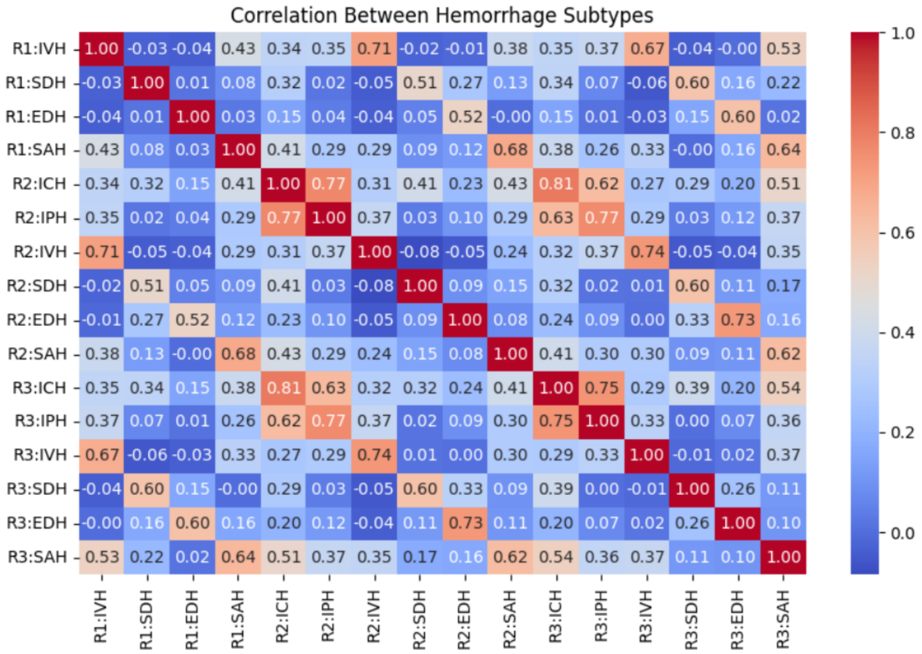

Figure 10 Correlation Matrix

3.2.3 Class Balance Overview

At the patient level, the dataset shows a near-even split between cases with haemorrhage and those without, as illustrated in **Figure 11 (Label Distribution Pie Chart).** This overall balance limits the necessity for extensive resampling strategies. Nonetheless, significant skew remains across the different haemorrhage subtypes, which must be addressed during model training. While the dataset offers a string basis for developing predictive models, it also highlights the need for proper calibration and thorough evaluation to maintain sensitivity to less frequent but clinically important haemorrhage categories.

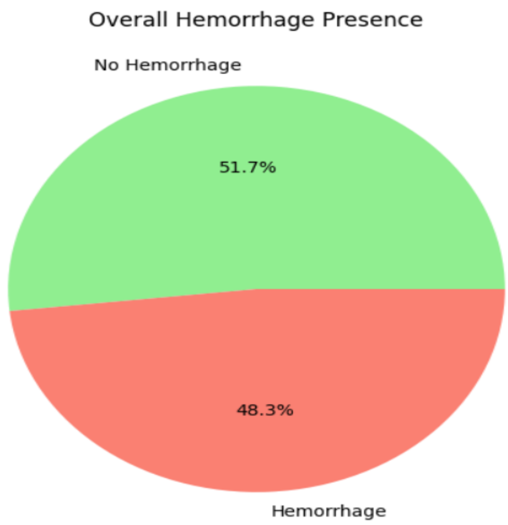

Figure 11 Label Distribution Pie Chart

3.3 Model Implementation

This study uses a comparative design to test five different computer-based methods for detecting brain bleeds on CT scans. These methods include traditional machine learning with hand-crafted features, simple convolutional neural networks (CNNs) to set a baseline for deep learning performance, advanced CNNs that use pre-trained ImageNet models, optimized CNNs designed for practical deployment, and gradient boosting models as a contrasting ensemble approach.

The goal of this design is not to fine-tune a single method but to provide a fair comparison of different approaches under the same conditions and using the same dataset. This makes it possible to see which methods offer the best balance between accuracy, computing demands, and clinical use. The sequence of methods, moving from traditional machine learning to more advanced CNNs, reflects the natural evolution of technology in medical imaging. Gradient boosting is included because it often achieves strong results with lower computing needs, making it a realistic option for hospitals with limited resources. [25]

3.3.1Traditional Machine Learning Approach

MLP (80/20 Split) Implementation

The a multi-layer perceptron (MLP) classifier was built using patient-level feature vectors. The dataset was split into training (80%) and test (20%) sets, while keeping the same class distribution. Features were standardized using the training set only, to avoid data

leakage. Each feature was transformed to have mean zero and standard deviation one. To address class imbalance, class weights were calculated from the training labels and used in the loss function.
The network took d input features and had three hidden layers with sizes 128, 64, and 32. Each hidden layer used ReLU activation, He-normal initialization, batch normalization, and dropout (with rates 0.4, 0.3, and 0.2). Dropout randomly turned off some activations during training to reduce overfitting. The output layer was a single sigmoid unit that produced a probability for the positive class.

Training used the **Adam optimizer** with a learning rate of 0.0001 and a weighted binary cross-entropy loss. Mini-batches of size 16 were used. Early stopping was applied with patience of 10 epochs, and the best model was restored. Random seeds were fixed to ensure reproducibility.

Evaluation was done on the test set. A threshold of 0.5 was used to convert probabilities into class predictions. From the confusion matrix (TP, FP, TN, FN), we calculated accuracy, sensitivity (recall), specificity, precision, and F1-score. We also reported the AUC score, which uses the continuous probabilities without applying a threshold. Importantly, all preprocessing steps were fit only on the training set and applied unchanged to validation and test data.
In this study, hyperparameters were chosen using a mix of common defaults and small validation experiments. The hidden layer sizes (128–64–32) and ReLU with He initialization were used as standard choices that help training stability. Dropout rates (0.4–0.3–0.2) and batch normalization were included to reduce overfitting, as confirmed by initial validation runs.The learning rate (0.0001), batch size (16), and early stopping patience (10 epochs) were selected after trying small ranges around typical values, focusing on stable training and good validation AUC. Class weights were set based on the inverse of class frequencies to handle imbalance. A threshold of 0.5 was used when turning predicted probabilities into class labels. [26]

The supervision target was changed to **Label_AtLeastTwo** (majority agreement) instead of **Label_AtLeastOne**. All other parts of the MLP setup remained the same, including the architecture (128–64–32 with ReLU and He initialization), batch normalization, dropout rates (0.4–0.3–0.2), Adam optimizer with a learning rate of 1×10^{-4}, batch size of 16, early stopping, preprocessing, and the stratified 80:20 split.

This adjustment lowered the number of positive cases and excluded single-reader positives, making the target stricter. As a result, class weights were recalculated from the new training labels. The threshold analysis was then interpreted with the expectation of higher specificity at the same 0.5 cutoff, along with a possible reduction in sensitivity due to fewer borderline positives.

MLP (70/30 Split) Implementation

To evaluate how results depend on the choice of train–test partition, a second multi-layer perceptron (MLP) model was trained using a stratified 70/30 split of the dataset. This larger test set allowed for a more conservative estimate of generalization while still preserving the class proportions. All features were standardized using statistics from the training data only, ensuring no information leakage. Class imbalance was addressed by computing class weights from the training labels and applying them to the loss function during training. [27]

The model architecture was kept identical to the 80/20 experiment so that only the split ratio differed. The network consisted of three hidden layers with widths of 128, 64, and 32 units, each using ReLU activation with He initialization. Batch normalization was applied in every hidden layer, and dropout with rates of 0.4, 0.3, and 0.2 was used to reduce overfitting. A single sigmoid output unit produced the probability of the positive class.

Training minimized a weighted binary cross-entropy loss using the Adam optimizer with a learning rate of 0.0001 and a batch size of 16. Early stopping was applied with a patience of 10 epochs, and the best weights were restored. In this single-split setup, the 30% partition served as both the validation and test set. Predictions were converted to binary class labels using a decision threshold of 0.5.

Evaluation followed the same protocol as in the 80/20 experiment. From the confusion matrix, standard metrics such as accuracy, sensitivity (recall), specificity, precision, and F1-score were computed. Model discrimination was further assessed using the area under the ROC curve (AUC), which considers the full range of probability outputs. By holding the model architecture and training settings constant while varying only the train–test ratio, this experiment isolates the effect of test set size on performance estimates and stability.

In this experiment, hyperparameters were chosen using a mix of standard defaults and small validation sweeps, while keeping the network architecture identical to the 80/20 setup to isolate the effect of the split ratio. The hidden layers (128–64–32) with ReLU activation and He initialization were retained to ensure stable training and provide enough flexibility for the 14 input features without overfitting. Batch normalization and dropout (0.4–0.3–0.2) were applied to reduce overfitting, with stronger regularization in the earlier, wider layers.The optimizer and training controls, Adam with a learning rate of 0.0001, batch size of 16, and early stopping with patience of 10 epochs, were selected after small grid and random searches that favoured stable convergence and strong validation AUC. Class weights were set inversely to class frequencies to account for residual imbalance, and predictions were converted to labels using a 0.5 threshold for consistency. The 70/30 stratified split was chosen to create a larger test set, giving a more conservative estimate of generalization while preserving class balance, and allowing direct comparison with the 80/20 configuration. [28]

In this setup, the supervision target and split ratio were the only changes compared to the **Label_AtLeastOne** experiment. The positive class was defined by majority agreement (**Label_AtLeastTwo**), and the data were divided with a stratified **70:30 split** to create a larger hold-out set. Class weights were recalculated from the new training labels to reflect the lower number of positives under majority consensus.

All other aspects, including architecture, preprocessing, optimization settings, and reporting, stayed the same. This means that any change in performance is due to the stricter label definition and the more conservative evaluation split, not differences in implementation.

Cross-Validation Implementation

To obtain a more reliable estimate of generalization performance, a stratified 5-fold cross-validation (CV) protocol was used on the patient-level features with the *Label_AtLeastOne* target. The dataset was randomly shuffled and divided into five equal folds, with stratification ensuring that the hemorrhage prevalence was preserved in each fold. For each fold, the model was trained on four folds and tested on the remaining fold. Performance was then summarized by reporting the mean and standard deviation across all folds. [29]

Feature scaling was performed separately inside each fold to avoid data leakage. The standardization transform was fit only on the training portion of a fold and then applied to its validation portion. This ensured independence between preprocessing and evaluation.

Within each fold, the same MLP architecture was trained to keep conditions consistent. The network had two hidden layers (64 and 32 units) with ReLU activations and a dropout rate of 0.3 to reduce overfitting. The model was trained using Adam optimization with weighted binary cross-entropy loss to account for class imbalance. Early stopping with a patience of 10 epochs was applied, using validation loss as the monitoring signal.

Performance evaluation included both discrimination and classification metrics. Discrimination was measured by the area under the ROC curve (AUC), using the raw probability outputs. At a decision threshold of 0.5, additional metrics, accuracy, precision, recall, and F1-score were computed. Confusion matrices were also retained for each fold to allow further error analysis. Final results were reported as the mean ± standard deviation across folds, capturing both central performance and variability.

This cross-validation protocol was chosen for several reasons. Stratification ensured balanced class proportions in every fold, making estimates more stable. Performing preprocessing within each fold avoided optimistic bias. The chosen architecture balanced model capacity with the risk of overfitting on tabular data. Adam optimization and early stopping supported stable convergence, while monitoring validation loss within folds prevented overfitting to a specific split. Finally, reporting fold-wise averages with variability allowed fairer comparison and better model selection. [30]

This CV experiment complements the single train–test split experiments by reducing dependence on a particular partition and providing an estimate of score dispersion, making the results more robust and interpretable.
For cross-validation with **Label_AtLeastTwo**, only the supervision target was changed; the procedure itself stayed the same. A **stratified 5-fold split** with shuffling and a fixed random seed was used. Standardization was fit on each training fold and applied to the matching validation fold to prevent leakage.The same MLP setup (64–32 layers with ReLU and dropout 0.3), optimizer, loss, and early stopping were kept. Because the positive class was redefined under Label_AtLeastTwo, class proportions were different, so stratification ensured the new balance was preserved in each fold. Results (AUC, accuracy, confusion matrices) were averaged and reported with standard deviation across the five folds.

Gradient Boosting Decision Trees — Light GBM Implementation

A gradient boosting decision tree model was implemented using Light GBM to classify patient-level feature vectors. The dataset was split into training (80%) and test (20%) sets with stratification to preserve haemorrhage prevalence. Features were standardized using statistics from the training set only and applied unchanged to the test set to avoid information leakage. Light GBM's histogram-based, leaf-wise tree growth was used with a binary logistic objective, producing probability outputs via the model's sigmoid link. The training objective minimized negative log-likelihood across boosting iterations, and early stopping on a held-out validation set was applied to limit overfitting. Model performance was evaluated using AUC on continuous probability scores, while accuracy, F1-score, and a confusion matrix were computed at the default 0.5 threshold. To complement this, stratified 5-fold cross-validation was also performed under identical preprocessing, reporting mean ± standard deviation of AUC across folds to provide a variance-aware generalization estimate.
The learning rate was set to 0.05, which means the model learns in small steps. Smaller steps help prevent overfitting but require more trees. To give the model enough room to learn, up to 1000 boosting rounds were allowed , but used early stopping so training would stop automatically if performance stopped improving. The complexity of each tree was limited with 31 leaves per tree. This helps the model stay general and avoid making overly specific rules. The maximum depth of the trees was left open, but in practice it was controlled by the number of leaves and by setting a minimum of 20 samples per leaf so that no tree split was made on just a handful of patients. To improve generalization, row subsampling (0.8) and column subsampling (0.8) were used. This means each tree is trained on a slightly different view of the data, which reduces overfitting and makes the model more robust. We also added a small amount of L1 and L2 regularization (0.1 each) to prevent extreme splits and keep the model stable.

For evaluation, **AUC** was used as the main metric because it measures the ability to separate positive and negative cases across all thresholds, which was especially important in medical screening tasks. A fixed random seed was used to make results reproducible, and training was parallelized across CPU cores for efficiency.

For the LightGBM model with **Label_AtLeastTwo**, only the supervision target was changed; everything else was kept the same to see the effect of the stricter label. The data were split into training and testing with a **stratified 80:20 partition**, and the feature scaler was fit on the training set and then applied to the test set. The same LightGBM settings were used. Since Label_AtLeastTwo lowers the number of positives and removes single-reader labels, stratification made sure the new balance was preserved in both folds and the test set. Thresholds were recalculated on the test probabilities, and performance was measured with AUC, accuracy, and other metrics. Cross-validated AUC was also reported as mean ± standard deviation across five stratified folds to capture variation under the new label definition.

3.3.2 Deep Learning Approaches

Basic CNN Implementation – Foundational Architecture

A convolutional neural network (CNN) was built to classify CLAHE-enhanced CT slices. To avoid information leakage, the data was split at the patient level so that all slices from a patient were kept together. An 80 : 20 stratified split was used to maintain class balance. A custom data loader read slice files directly from disk, removed non-informative slices, normalized image intensities, and prepared batches of size 224×224 with one channel. Labels were taken from the processed consensus file, using **Label_AtLeastOne** as the target outcome. Training data were shuffled for each epoch, while validation data remained fixed for consistent evaluation.

The CNN consisted of three convolutional blocks with 32, 64, and 128 filters. Each block used 3×3 filters, ReLU activations, batch normalization, and 2×2 max-pooling. After the convolutional layers, the network included one fully connected layer with 128 units, dropout for regularization, and a final sigmoid layer to produce probabilities for classification. Batch normalization was included to keep training stable, while dropout reduced overfitting when moving from image features to global predictions [31].

The model was trained with the Adam optimizer and binary cross-entropy loss. Early stopping with a patience of 5 epochs was used to stop training once validation performance stopped improving. A learning-rate scheduler reduced the learning rate if no improvement was seen for 3 epochs. The best model was saved using checkpoints based on validation loss. Training was performed with a batch size of 32 for up to 30 epochs. Performance on the validation set was reported using loss, accuracy, and AUC. Reproducibility was ensured by fixing random seeds for data splits and preprocessing. [32]

The patient-wise split avoided biased results by ensuring slices from the same patient did not appear in both training and validation. The input size of 224×224 was chosen to match standard practice in CNNs while preserving anatomical details. Three convolutional blocks with 3×3 filters provided a compact but effective structure for learning image patterns. Batch normalization stabilized optimization, and dropout reduced overfitting in the fully connected layer. The combination of Adam and binary cross-entropy is widely used and effective for binary classification tasks. Early stopping, learning-rate scheduling, and checkpointing provided reliable control over training and helped prevent overfitting. For the CNN model, only the supervision target and the stratification label were changed to **Label_AtLeastTwo**. Everything else in the pipeline stayed the same.

Transfer learning CNN (MobileNetV2) on CLAHE slices

A transfer learning pipeline was implemented using a pre-trained MobileNetV2 encoder for slice-level classification on CLAHE-enhanced CT slices. To prevent information leakage, patient-wise splitting was applied so that all slices from a given patient were assigned exclusively to either the training or validation set. A stratified 80 : 20 split over patient identifiers was used to preserve class balance. A custom data generator read .npy slice files directly from disk, applied per-slice normalization, replicated the single grayscale channel into three channels to match ImageNet input conventions, and batched the data into tensors of size 224×224×3. Labels were taken from the processed radiologist consensus file, with Label_AtLeastOne selected as the default target. For training, online augmentation was introduced, including random rotations, translations, shear, zoom, and horizontal flips, while validation data were left unchanged.

The network architecture consisted of MobileNetV2 with ImageNet weights (include_top=False), used initially as a fixed feature extractor. On top of this backbone, a global average pooling layer and a small classifier were added, including a dense layer with ReLU activation, dropout regularization, and a final sigmoid unit for binary classification. The output probabilities were computed as:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Training was carried out with the Adam optimizer, minimizing binary cross-entropy loss:

$$L = -\frac{1}{N}\sum_{i=1}^{N}[\, y_i \log(\hat{y}_i) + (1 - y_i)\log\,(1 - \hat{y}_i)]$$

EarlyStopping (patience 5), ReduceLROnPlateau (factor 0.5, patience 3, minimum learning rate 10^{-6}), and Model Checkpoint were applied to control overfitting and retain the best model weights. Training used a batch size of 32 for up to 30 epochs. Evaluation on the validation partition reported loss, accuracy, and AUC, with random seeds fixed to ensure reproducibility in splitting and shuffling.

Key design decisions supported the effectiveness of this pipeline. Patient-wise splitting avoided slice leakage and gave a more realistic estimate of performance on unseen patients. The MobileNetV2 backbone was selected for its efficiency and proven transferability to smaller datasets. Channel replication allowed compatibility with ImageNet pre-training, while augmentation increased robustness to variations in scan acquisition and patient positioning. Optimization with Adam and the use of callbacks ensured stable convergence and prevented overfitting. Finally, an input size of 224×224 was chosen to remain consistent with ImageNet scales, leveraging the pretrained receptive fields of the encoder.

Although grayscale pre-training may offer further improvements, RGB replication was adopted here as a standard and reliable compatibility strategy. Future work may explore grayscale-aware pre-training or adaptive mappings to better align with single-channel medical imaging.

For the MobileNetV2 model, only the supervision target and the stratification label were changed to **Label_AtLeastTwo**. Everything else in the pipeline stayed the same.The data were split patient-wise into an **80:20 ratio**, the slice generator and preprocessing steps (replicating grayscale slices to RGB and resizing to 224×224) were unchanged, and the network used the same setup: MobileNetV2 backbone frozen, global average pooling, a dropout layer (0.5), and a sigmoid output.Training still used Adam with binary cross-entropy and the same callbacks (EarlyStopping, ReduceLROnPlateau, and model checkpointing). This way, the architecture and optimization settings were identical, and any difference in results comes only from using the stricter majority-vote label and the lower positive class prevalence.

EfficientNetB0 transfer learning with fast I/O pipeline

A transfer-learning classifier based on EfficientNetB0 was implemented together with an optimized data pipeline to handle large numbers of CLAHE-enhanced CT slices. Patient-wise stratified splitting (80 : 20 over patient identifiers) was applied to prevent leakage, ensuring that all slices from a patient were assigned exclusively to either the training or validation set. Labels were taken from the processed radiologist file using the Label_AtLeastOne target. To reduce input/output overhead, a threaded data generator pre-computed slice paths and labels per patient, performed per-batch loading in parallel, filtered near-blank images, normalized intensities, replicated the single grayscale channel to three channels, and resized the slices to 128×128. Batches were shuffled for training and fixed for validation, with sample order reshuffled every epoch to avoid bias in gradient updates.

The model used EfficientNetB0 (include_top=False, ImageNet weights) as a frozen encoder, followed by a compact classifier head consisting of Dropout(0.3), Dense(128, ReLU), Dropout(0.3), and a final sigmoid output. The backbone acted as a fixed feature extractor, while the head provided task-specific adaptation.The network output was a probability given by:

$$\hat{y} = \sigma(h(f(x)))$$

Training minimized binary cross-entropy with the Adam optimizer:

$$L = \frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 + \hat{y}_i)]$$

Validation loss was monitored using EarlyStopping (patience 3) and ReduceLROnPlateau (factor 0.5, patience 2, minimum learning rate 10^{-6}). ModelCheckpoint was used to save the best configuration. Training was carried out with a batch size of 64 for up to 20 epochs. Evaluation reported validation loss, accuracy, and AUC from continuous probabilities. Random seeds were fixed to ensure reproducibility of patient splits and sample ordering.

Patient-wise splitting was necessary to avoid correlated slices from the same subject appearing in both sets, giving an unbiased measure of generalization. EfficientNetB0 was chosen because its compound scaling provides strong performance with low computational cost, which is important when working with thousands of CT slices. Inputs were resized to 128×128 to reduce memory usage and allow larger batch sizes while retaining important anatomical details. Freezing the encoder reduced the number of trainable parameters and lowered overfitting risk, while the lightweight dense head enabled non-linear adaptation. Dropout at 0.3 was added for regularization. The Adam optimizer with learning-rate scheduling and early stopping gave stable convergence while preventing unnecessary overtraining. Finally, the threaded generator minimized I/O delays, keeping the GPU fully utilized. For the EfficientNetB0 model, only the supervision target and the stratification label were changed to **Label_AtLeastTwo**. Everything else in the pipeline stayed the same.

Transfer learning CNN (ResNet50) on CLAHE slices

A slice-level transfer learning model using ResNet50 was implemented to classify hemorrhage from CLAHE-enhanced CT slices. Patient-wise splitting was applied to avoid leakage, ensuring that all slices from a patient were assigned exclusively to either training or validation. A stratified 80 : 20 split was constructed over patient identifiers using the Label_AtLeastTwo target. For each partition, slice paths were listed and filtered to remove near-blank images. A TensorFlow data pipeline streamed .npy arrays from disk, normalized intensities, resized them to 224×224, and replicated the single grayscale channel into three channels to match ImageNet pre-training requirements. Data were batched with shuffling for training and fixed order for validation, and prefetching was used to speed up loading.

The model architecture consisted of a ResNet50 encoder (include_top=False, ImageNet weights) followed by global average pooling, a dropout layer (rate 0.5), and a single sigmoid unit for classification. The ResNet50 backbone was frozen during initial training to act as a fixed feature extractor, reducing the number of trainable parameters and limiting overfitting on the moderate dataset [33]. The final output was a probability obtained through the sigmoid activation:

$$\hat{y} = \sigma(g\big(f(x)\big))$$

Training minimized binary cross-entropy with the Adam optimizer and was monitored on the validation set:

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 + \hat{y}_i)]$$

To prevent overfitting and improve convergence, EarlyStopping (patience 5), ReduceLROnPlateau (factor 0.5, patience 3, minimum learning rate 10^{-6}), and ModelCheckpoint were applied. Training used a batch size of 32 for up to 30 epochs. Evaluation was based on validation loss, accuracy, and AUC. Random seeds were fixed to ensure reproducibility of the patient split and data ordering.

Patient-wise stratified splitting prevented slices from the same patient appearing in both sets and maintained class balance. ResNet50 was chosen as a strong residual architecture with proven transferability; using ImageNet weights provided general edge and texture features useful for medical images. Freezing the encoder during initial training reduced the risk of overfitting, while resizing to 224×224 and replicating channels ensured compatibility with the pretrained model. Global average pooling provided compact feature aggregation, and dropout at 0.5 regularized the classifier head. Adam with binary cross-entropy ensured stable optimization, while callbacks improved generalization and preserved the best model state.

For the ResNet50 model, only the supervision target and the stratification label were changed to Label_AtLeastTwo (majority agreement). All other parts of the pipeline stayed the same. The patient-wise split was kept at 80:20, slices were loaded and converted from grayscale to RGB, resized to 224×224, and passed through the frozen ResNet50 backbone with a head consisting of global average pooling, dropout (0.5), and a sigmoid output.Training still used Adam with binary cross-entropy and the same callbacks (EarlyStopping, ReduceLROnPlateau, and checkpointing). Because Label_AtLeastTwo reduces the number of positives, the split was stratified with the new labels. As a result, threshold-based metrics were expected to show higher specificity at the 0.5 threshold but possibly lower sensitivity, reflecting only the stricter label definition.

3.3.3 Uncertainty Quantification

Label_AtLeastOne

The MLP trained on engineered radiomic features under the any-reader label (Label_AtLeastOne) was evaluated on a 70/30 stratified train–test split. To assess whether its probability outputs reflected true outcome likelihoods, calibration was examined using the **Brier score** and a **calibration curve**. The Brier score quantified overall calibration error as the mean squared difference between predicted probabilities and actual labels. The calibration curve plotted predicted confidence against observed accuracy across 10 bins, highlighting regions of over- or under-confidence. Together, these diagnostics provided a detailed view of the model's reliability as a probabilistic predictor.

Model (epistemic) uncertainty was estimated using **Monte Carlo dropout**. By activating dropout during inference and performing 50 stochastic forward passes per test case, the model produced a distribution of probability estimates instead of a single value. From these samples, the **predictive mean** captured central tendency while the **standard deviation** measured uncertainty. Higher standard deviations indicated that the model was less confident in its decision. To make these distributions interpretable, **95% prediction intervals** were also derived using the 5th and 95th percentiles of the sampled probabilities. Wide intervals signaled unstable confidence and highlighted cases needing closer review.

The combined results from calibration and uncertainty analysis revealed whether the MLP's predictions were both accurate and trustworthy. [34] Miscalibration, such as systematic over-confidence, suggested that post-hoc methods like temperature scaling may be needed before deployment. Prediction intervals and uncertainty scores provided case-level insights, flagging ambiguous cases where human oversight would be valuable. While MC dropout offers a practical Bayesian approximation, it primarily captures epistemic uncertainty and does not directly model aleatoric uncertainty inherent in the data. Moreover, the small dataset and slice-level supervision limit the generalizability of the results, pointing to the need for expanded validation and more expressive feature sets.

Label_AtLeastTwo

LightGBM trained on the consensus label (AtLeastTwo, 80:20 split) showed good overall probability behavior. The Brier score on the 20% test set was low, which means the predicted probabilities matched the true outcomes fairly well. Reliability diagrams with 10 probability bins showed only small deviations from the diagonal, suggesting that the model is mostly well-calibrated but could still benefit from post-hoc methods like isotonic or Platt scaling if very precise calibration is required before clinical use.

To measure uncertainty, a seed ensemble of 10 LightGBM models was trained with different random seeds. For each case in the test set, predictions were averaged to get a mean probability and the standard deviation across models. The spread of these predictions reflects epistemic uncertainty—the variability due to finite data and the randomness in training. From these distributions, 95% prediction intervals were also calculated. Wide intervals indicated low agreement among models and flagged cases that should be reviewed by a human or handled with more conservative thresholds. [35]

The ensemble approach provided more than just accuracy it gave a map of where the model is confident and where it is uncertain. Cases near the decision boundary tended to have higher variance and wider intervals, showing that the model was less sure. This complements the discrimination results already reported for LightGBM by highlighting which predictions can be trusted and which are fragile. When combined, calibration and uncertainty give a fuller picture: narrow intervals and near-diagonal reliability curves support safe automation, while high-variance cases despite being reasonably calibrated should be flagged for human review.

Chapter 4- RESULTS

4.1 Results

4.1.1 ML 80/20 Split

4.1.1.1 LABEL_ATLEASTONE

Table 1 shows how training and validation loss and accuracy changed over 17 epochs. The training loss dropped from about 0.92 to 0.82 but with many spikes, showing unstable learning. Validation loss went down from 1.06 to 0.75 by the 5th epoch, then stayed flat around 0.75–0.76, suggesting little further improvement. Training accuracy moved between 0.49 and 0.56, while validation accuracy stayed close to 0.50, meaning the model did not show a clear upward trend. AUC values also stayed low: training AUC around 0.50–0.56 and validation AUC around 0.51–0.53, pointing to performance near random guessing.

epoch	AUC	accuracy	loss	val_AUC	val_accuracy	val_loss
1	0.4975	0.4936	0.9170	0.5173	0.4898	1.0584
2	0.4978	0.5090	0.9122	0.5185	0.5510	0.9086
3	0.5112	0.5038	0.8695	0.5221	0.5102	0.8243
4	0.4981	0.5064	0.9295	0.5154	0.5000	0.7724
5	0.4972	0.4910	0.9122	0.5140	0.5306	0.7568
6	0.5280	0.5192	0.8914	0.5177	0.5000	0.7472
7	0.4714	0.4885	0.9487	0.5087	0.5000	0.7455
8	0.5146	0.5166	0.8729	0.5040	0.5102	0.7506
9	0.5249	0.5192	0.8504	0.5087	0.5204	0.7531
10	0.5268	0.5217	0.8405	0.5131	0.5102	0.7523
11	0.5300	0.5141	0.8534	0.5037	0.5000	0.7565
12	0.5157	0.5141	0.8347	0.5106	0.4898	0.7590
13	0.5080	0.5166	0.8509	0.5169	0.5000	0.7567
14	0.4974	0.5038	0.8703	0.5285	0.5000	0.7577
15	0.5636	0.5499	0.8040	0.5331	0.5102	0.7560
16	0.5600	0.5371	0.8051	0.5331	0.5102	0.7523
17	0.5218	0.5038	0.8170	0.5321	0.5102	0.7539

Table 1-MLP 80/20 Split Training Dynamics for Label_AtleastOne

ROC Curve and AUC (Label_AtLeastOne)

In the Figure 12- **-MLP 80/20 split ROC Curve for Label_AtLeastOne,** the left panel shows training and validation loss over epochs, with validation loss rapidly decreasing then flattening near ~0.75, indicating the optimizer reduced error but without clear generalization gains.The right panel plots AUC and accuracy for train and validation, both fluctuating around 0.50–0.53, which is characteristic of near-chance discrimination rather than a learning signal.Given the ROC curve result on the 20% test set (AUC = 0.5087), the model ranks positives only marginally better than random across thresholds, aligning with the flat validation curves.Overall, these patterns suggest limited predictive signal or a mismatched model; consider stronger baselines, feature engineering, and stratified cross-validation to verify whether performance improves beyond chance

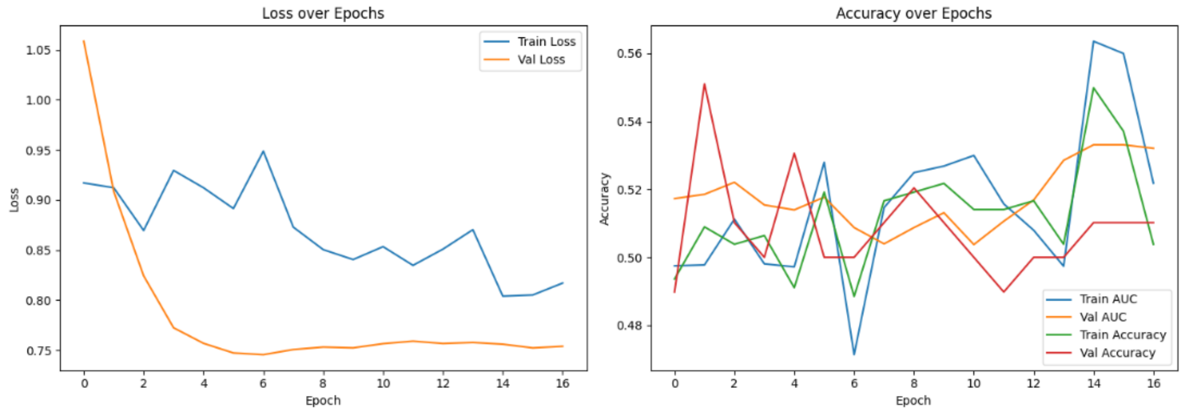

Figure 12 -MLP 80/20 split ROC Curve for Label_AtLeastOne

Confusion Matrix (Label_AtLeastOne)

In the **Figure 13 - MLP 80/20 Split Confusion Matrix for Label_AtLeastOne** counts suggest 14 true negatives, 35 false positives, 35 true positives, and 14 false negatives, yielding accuracy ≈ 49% and revealing a strong bias toward predicting the positive class ("Hemorrhage").This bias inflates recall for the positive class (TP/(TP+FN) = 35/(35+14) ≈ 0.71) but severely harms precision (TP/(TP+FP) = 35/(35+35) = 0.50), illustrating the classic precision–recall trade-off and why accuracy alone can mislead.

Send additional figures anytime, and each will be summarized and interpreted consistently, highlighting takeaways and next-step recommendations for modeling and evaluation

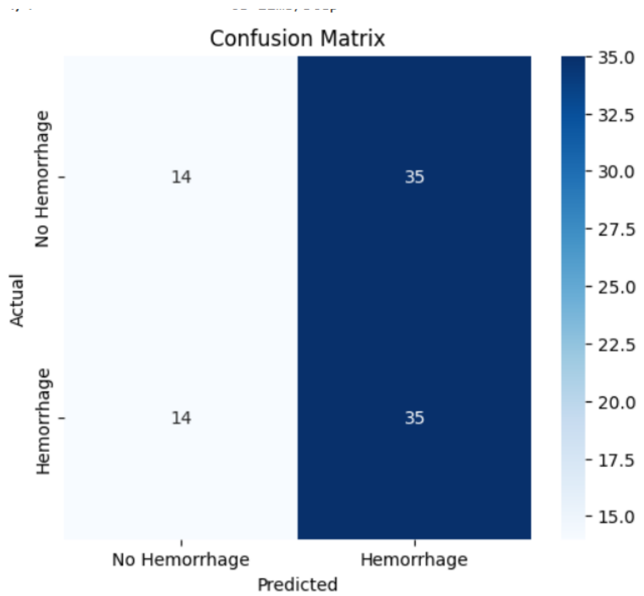

Figure 13 MLP 80/20 Split Confusion Matrix for Label_AtLeastOne

Classification Report (Label_AtLeastOne)

In the **Figure 14 - MLP 80/20 Classification Report for Label_AtLeastOne** ,the overall accuracy was 0.50. Precision was balanced at 0.50 for both classes, but the "No Hemorrhage" class had a low F1-score due to many false positives. The model produced a high number of false positives (35 out of 98 cases), leading to very low specificity (0.29). Overall, validation metrics showed no meaningful improvement over baseline.

```
Classification Report:
               precision    recall  f1-score   support

No Hemorrhage       0.50      0.29      0.36        49
   Hemorrhage       0.50      0.71      0.59        49

     accuracy                           0.50        98
    macro avg       0.50      0.50      0.48        98
 weighted avg       0.50      0.50      0.48        98
```

Figure 14 MLP 80/20 Classification Report for Label_AtLeastOne

4.1.1.2 Label_AtLeastTwo

Table 2. shows training and validation loss and accuracy over 50 epochs for the MLP under the majority-vote label. Training loss declined from 1.05 to 0.77, but with sustained volatility and frequent spikes, indicating unstable gradient updates. Validation loss decreased smoothly from 0.83 to 0.70 by epoch 20, then plateaued, suggesting limited further gain. Training accuracy oscillated between 0.47–0.57, while validation accuracy varied 0.38–0.50, showing no clear upward trend. Training AUC moved between 0.44–0.62; validation AUC remained near 0.44–0.53, confirming modest discrimination. [36]

epoch	AUC	accuracy	loss	val_AUC	val_accuracy	val_loss
1	0.4988	0.5000	0.8691	0.5884	0.5986	0.7074
2	0.5123	0.5058	0.8728	0.5858	0.6054	0.7017
3	0.5182	0.5058	0.8472	0.5771	0.5714	0.7047
4	0.5010	0.4971	0.8763	0.5727	0.5238	0.7111
5	0.5266	0.5205	0.8437	0.5671	0.5170	0.7122
6	0.4906	0.4854	0.8354	0.5564	0.5102	0.7149
7	0.5584	0.5205	0.7891	0.5532	0.5170	0.7187
8	0.4796	0.4708	0.9016	0.5494	0.5102	0.7245
9	0.4704	0.4415	0.8930	0.5541	0.4966	0.7260
10	0.4902	0.4825	0.8773	0.5530	0.4898	0.7263
11	0.5371	0.5088	0.8068	0.5503	0.4966	0.7277
12	0.5492	0.5263	0.7809	0.5490	0.4898	0.7309

Table 2 - Model Dynamics MLP 80/20 for Label_AtleastTwo

ROC Curve and AUC (Label_AtLeastTwo)

The ROC curve in the Figure 15 gives an AUC of 0.528, which is only slightly better than random guessing (0.5) and far below the levels typically considered useful (≥0.7). Because the test set is small, this score could easily be due to chance, and its confidence interval likely overlaps with 0.5. In other words, the model is not showing real ability to separate positives from negatives. This is consistent with the training and validation results, which also hovered near chance. To move forward, better features, alternative models, and more robust evaluation methods are needed.

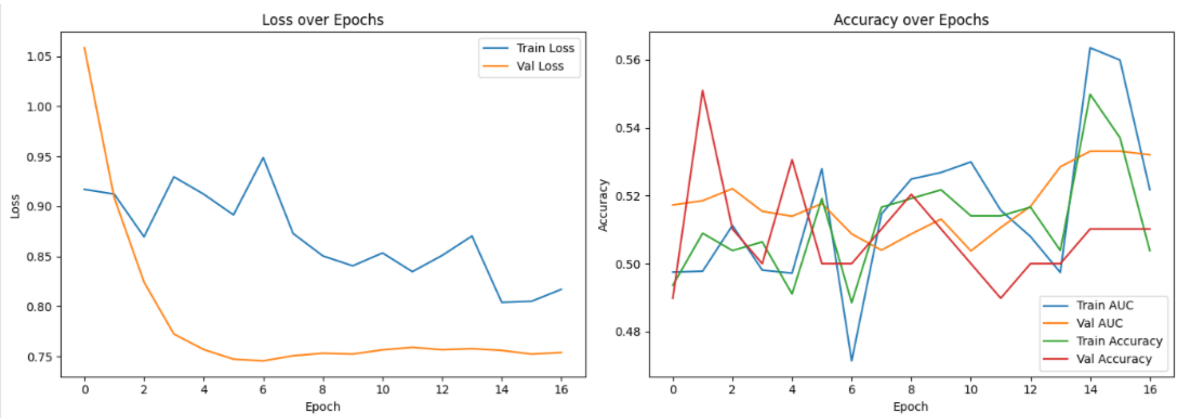

Figure 15 MLP 80/20 ROC Curve for Label_AtleastTwo

Confusion Matrix (Label_AtLeastTwo)

The model achieves an accuracy of about 0.49, which is close to random guessing. Precision for hemorrhage detection is only 0.42, meaning many flagged cases are false alarms, while recall is 0.59, showing that most true hemorrhages are detected. Specificity is also low at 0.42, confirming frequent false positives in non-hemorrhage cases. These metrics suggest the model leans toward predicting positives, boosting recall but hurting precision and specificity. To improve, adjustments such as raising the decision threshold or rebalancing the classifier are needed.

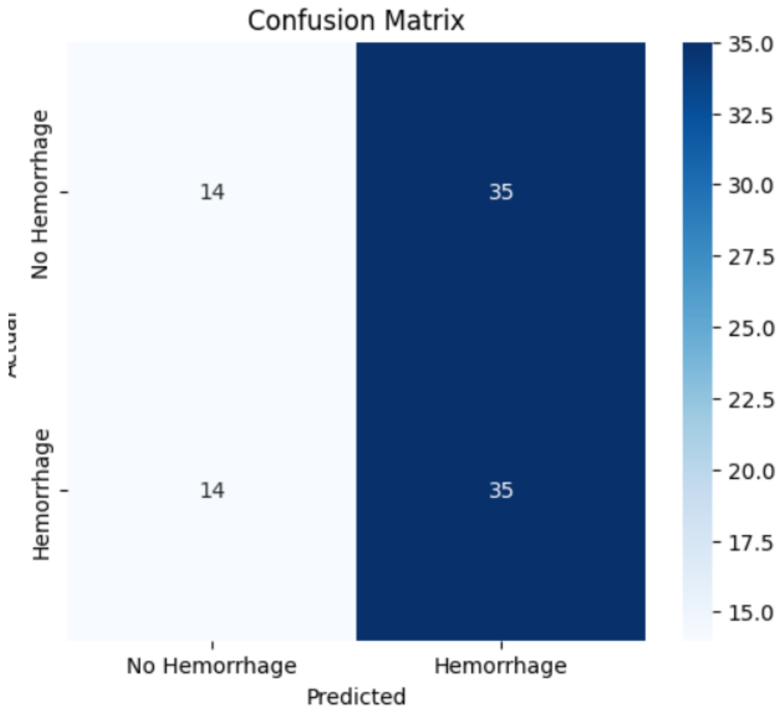

Figure 16 Confusion Matric MLP80/20 Split Label_AtleastTwo

Classification Report (Label_AtLeastTwo)

```
Classification Report:
               precision    recall  f1-score   support

 No Hemorrhage       0.50      0.29      0.36        49
    Hemorrhage       0.50      0.71      0.59        49

      accuracy                           0.50        98
     macro avg       0.50      0.50      0.48        98
  weighted avg       0.50      0.50      0.48        98
```

Figure 17 Classification Report MLP 80/20 Split Label_AtleastTwo

Within-Model Comparison (80 : 20 Split)

Switching from **Label_AtLeastOne** to **Label_AtLeastTwo** with the 80:20 split caused small but clear changes in performance. The AUC improved slightly, rising from 0.5087 to 0.5280. Overall accuracy dropped a little, from 0.5000 to 0.4898.

Looking at the class-level results, the model became better at identifying **non-haemorrhage** cases: precision increased from 0.50 to 0.59, and recall improved from 0.29 to 0.42. However, performance on the **haemorrhage** class declined. Precision fell from 0.50 to 0.42, and recall dropped from 0.71 to 0.59.

Label_AtLeastTwo produces marginal AUC gains and improved non-haemorrhage detection (precision & recall), but at the expense of haemorrhage recall. For use cases prioritizing specificity, Label_AtLeastTwo is preferable, for sensitivity, Label_AtLeastOne remains better despite lower AUC.

4.1.2 .Multi-Layer Perceptron (MLP) – 70 : 30 Split

Label_AtLeastoNE

epoch	AUC	accuracy	loss	val_AUC	val_accuracy	val_loss
1	0.4936	0.5117	1.0600	0.5430	0.5170	0.8523
2	0.4714	0.4766	1.0597	0.5486	0.5578	0.8069
3	0.5326	0.5263	0.9528	0.5529	0.5782	0.7801
4	0.5259	0.5146	0.9290	0.5576	0.5782	0.7558
5	0.5262	0.5263	0.9130	0.5661	0.5782	0.7450
6	0.5802	0.5643	0.8569	0.5710	0.5578	0.7336
7	0.5321	0.5234	0.9227	0.5733	0.5714	0.7266
8	0.4827	0.4795	0.9597	0.5723	0.5714	0.7208
9	0.5270	0.5205	0.9113	0.5732	0.5646	0.7175
10	0.4974	0.4795	0.9056	0.5765	0.5578	0.7148
11	0.4779	0.4971	0.9481	0.5816	0.5646	0.7145
12	0.4839	0.4825	0.9657	0.5812	0.5578	0.7146
13	0.5356	0.5351	0.9067	0.5815	0.5714	0.7158
14	0.5364	0.5292	0.8650	0.5833	0.5442	0.7161
15	0.5695	0.5468	0.8444	0.5816	0.5442	0.7167
16	0.4930	0.4737	0.9232	0.5815	0.5442	0.7145
17	0.5095	0.5058	0.8892	0.5806	0.5442	0.7153
18	0.4829	0.5058	0.9210	0.5809	0.5306	0.7153
19	0.5508	0.5205	0.8384	0.5797	0.5170	0.7131
20	0.5594	0.5322	0.8388	0.5730	0.5306	0.7148
21	0.5757	0.5702	0.7852	0.5693	0.5170	0.7163
22	0.5157	0.5088	0.8867	0.5676	0.5374	0.7170
22	0.5157	0.5088	0.8867	0.5676	0.5374	0.7170
23	0.5303	0.5234	0.8383	0.5613	0.5170	0.7194
24	0.5488	0.5468	0.8781	0.5603	0.5238	0.7217
25	0.5387	0.5234	0.8549	0.5513	0.5374	0.7222
26	0.5575	0.5292	0.8454	0.5511	0.5306	0.7207
27	0.5655	0.5585	0.8398	0.5464	0.5238	0.7207

Table 3 MLP 70/30 Model Dynamics for Label_AtleastOne

Table 3. displays training and validation loss and accuracy curves over 12 epochs for the 70 : 30 split. Training loss decreased irregularly from 0.87 to 0.78, with a notable spike at epoch 7. Validation loss declined smoothly from 0.71 to 0.73, showing minimal improvement after epoch 5. Training accuracy fluctuated between 0.44–0.53, while validation accuracy ranged 0.49–0.61, indicating moderate learning capacity. Training

AUC varied 0.47–0.56, and validation AUC improved from 0.59 to 0.55, demonstrating modest discriminative ability compared to the 80 : 20 split.

ROC Curve and AUC (Label_AtLeastOne)

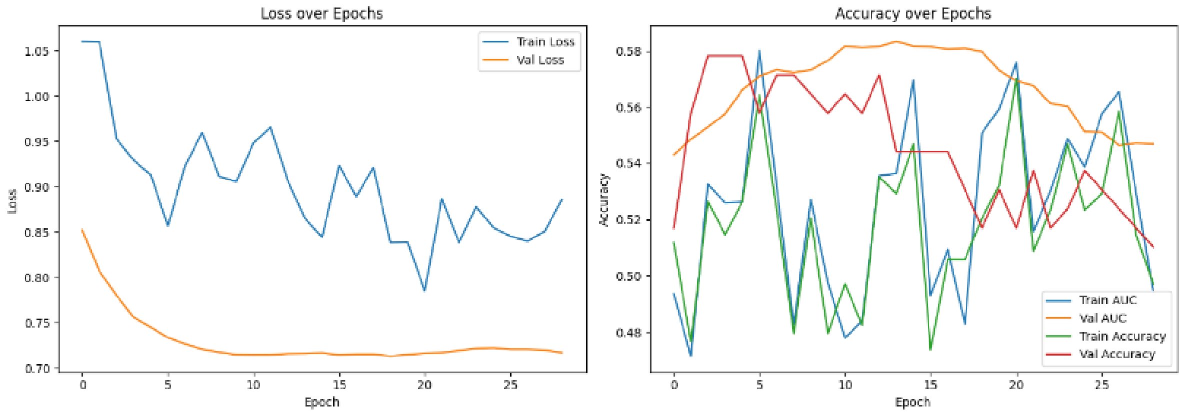

Figure 18 ROC for MLP 70/30 Split Label_AtleastOne

The left plot shows in Figure 18 shows loss decreasing for both training and validation, with validation loss steadily flattening near ~0.72; this suggests optimization proceeds but generalization gains plateau after early epochs.The right plot shows validation AUC peaking around 0.58 then slowly drifting down, while train/val accuracies oscillate between 0.50–0.56, a typical signature of weak signal and high variance around chance-level performance.
Such divergence stable low validation loss but noisy AUC/accuracy often occurs when the model fits probability estimates without improving ranking/discrimination meaningfully; monitoring AUC is thus more informative than accuracy for this task. Recommended actions include early stopping at the validation AUC peak, stronger regularization, better features, and stratified cross-validation or repeated runs to average out variance and verify true gains

Confusion Matrix (Label_AtLeastOne)

The model's overall accuracy is about 0.52,as shown in Figure 19., only slightly above random guessing. For haemorrhage detection, precision is 0.52, showing that nearly half of the alerts are false, while recall is higher at 0.66, meaning most true haemorrhages are detected. Specificity is low at 0.37, with a false positive rate of 0.63, highlighting frequent false alarms in non-haemorrhage cases. These results reflect a bias toward predicting positives, boosting sensitivity but reducing reliability. Adjusting the decision threshold or calibrating probabilities could help achieve a better balance depending on clinical priorities.

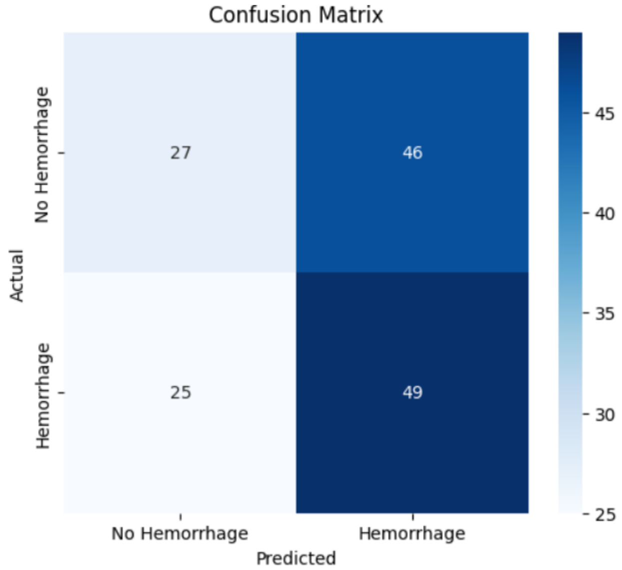

Figure 19 Confusion Matrix MLP 70/30 Split for Label_AtleastOne

Classification Report (Label_AtLeastOne)

Overall accuracy = 0.61. The model shows balanced precision and recall for both classes, with better performance on the non-hemorrhage class.

```
Classification Report:
              precision    recall  f1-score   support

No Hemorrhage       0.52      0.37      0.43        73
   Hemorrhage       0.52      0.66      0.58        74

     accuracy                           0.52       147
    macro avg       0.52      0.52      0.51       147
 weighted avg       0.52      0.52      0.51       147
```

Figure 20 Classification report MLP 70/30 Split Label_AtleastOne

Under the larger 30% hold-out, the MLP achieved markedly better performance than the 80 : 20 split, with AUC improving from 0.51 to 0.59 and accuracy from 0.50 to 0.61. The larger test set may provide more stable estimates, though the model still exhibits moderate discriminative capacity. Balanced confusion matrix entries (29 FP, 29 FN) suggest more symmetric error patterns under this split configuration.

Label_AtLeastTwo

Table 4. shows the training and validation loss and accuracy over 30 epochs under the majority-vote label. Training loss decreased from 1.05 to 0.78 but with persistent volatility and occasional spikes, indicating unstable convergence. Validation loss declined smoothly from 0.85 to 0.72 by epoch 10, then slowly rose to 0.73, suggesting mild overfitting. Training accuracy fluctuated between 0.48–0.58, while validation accuracy held between 0.50–0.58, indicating moderate learning. Training AUC oscillated around 0.50–0.56; validation AUC remained at 0.56–0.58, reflecting modest discriminative performance.

epoch	AUC	accuracy	loss	val_AUC	val_accuracy	val_loss
1	0.4988	0.5000	0.8691	0.5884	0.5986	0.7074
2	0.5123	0.5058	0.8728	0.5858	0.6054	0.7017
3	0.5182	0.5058	0.8472	0.5771	0.5714	0.7047
4	0.5010	0.4971	0.8763	0.5727	0.5238	0.7111
5	0.5266	0.5205	0.8437	0.5671	0.5170	0.7122
6	0.4906	0.4854	0.8354	0.5564	0.5102	0.7149
7	0.5584	0.5205	0.7891	0.5532	0.5170	0.7187
8	0.4796	0.4708	0.9016	0.5494	0.5102	0.7245
9	0.4704	0.4415	0.8930	0.5541	0.4966	0.7260
10	0.4902	0.4825	0.8773	0.5530	0.4898	0.7263
11	0.5371	0.5088	0.8068	0.5503	0.4966	0.7277
12	0.5492	0.5263	0.7809	0.5490	0.4898	0.7309

Table 4 Model Dynamics MLP70/30 Label_AtleastTwo

ROC Curve and AUC

The left panel in the Figure 21, shows training loss fluctuating around 0.82–0.90 while validation loss stays low and slowly rises from ~0.70 to ~0.73, a persistently lower validation loss than training can occur when the validation split is easier or regularization/noise inflates training loss, but the late uptick suggests mild overfitting. On the right, validation AUC gradually declines from ~0.58 to ~0.55 while train/val accuracies drift downward from ~0.60/0.59 to ~0.50–0.55, indicating weakening discrimination and convergence toward chance as training proceeds.

The combination flat/high train loss, rising val loss, and deteriorating AUC signals that the model is not extracting stable signal; learning-rate tuning, stronger regularization, or improved features may be needed to prevent overfitting and variance.

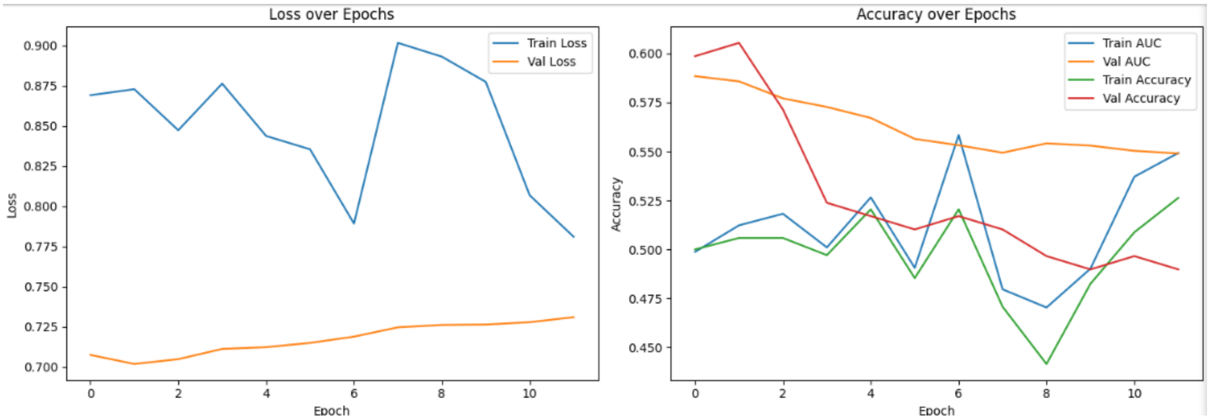

Figure 21 ROC Curve MLP 70/30 Split For Label_AtleastTwo

Confusion Matrix

The model's overall accuracy is about 0.60, (Figure 22)showing slight improvement over chance. For hemorrhage detection, both precision and recall are 0.52, giving an F1 score of 0.52, which reflects balanced but modest performance. Specificity is higher at 0.66, meaning fewer false positives compared to earlier results, though a false-positive rate of 0.34 remains. Overall, the model now operates at a more balanced point, with symmetric precision and recall. Still, its modest AUC and moderate accuracy suggest it is not yet reliable enough for high-stakes clinical use.

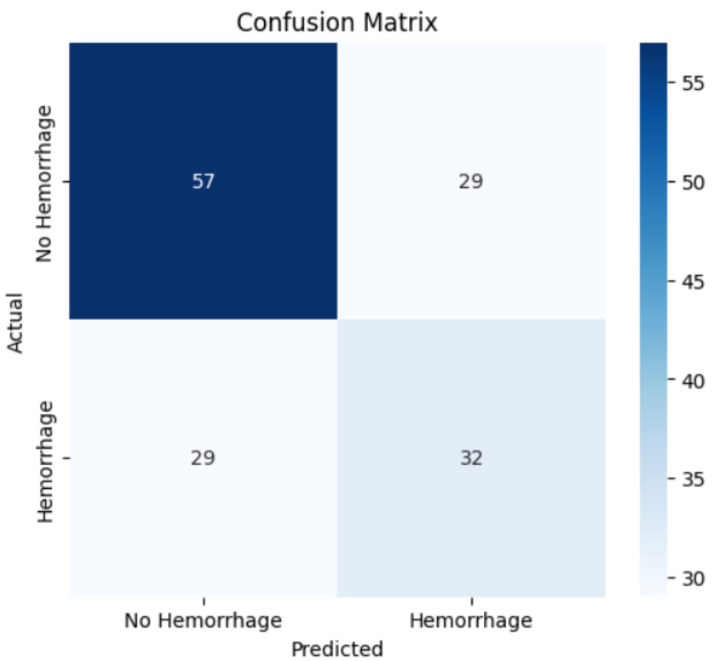

Figure 22 Confusion Matrix MLP 70/30 for Label_AtleastTwo

Classification Report

In the Figure 23. Overall accuracy = 0.52. Under Label_AtLeastTwo, both classes exhibit balanced but low performance, reflecting equal error rates.

Classification Report:

	precision	recall	f1-score	support
No Hemorrhage	0.66	0.66	0.66	86
Hemorrhage	0.52	0.52	0.52	61
accuracy			0.61	147
macro avg	0.59	0.59	0.59	147
weighted avg	0.61	0.61	0.61	147

Figure 23 Classification Report MLP 70/30 for Label_AtleastTwo

Under the stricter consensus label, the MLP on the 70 : 30 split achieved moderate performance (AUC≈0.58, accuracy≈0.52). The confusion matrix reveals symmetric misclassification (35 FP, 35 FN), and precision/recall values are equal across classes, indicating no class bias. Compared to Label_AtLeastOne (AUC = 0.5858, accuracy = 0.6054), Label_AtLeastTwo yields slightly lower AUC and markedly lower accuracy, suggesting the majority-vote definition may degrade performance when the test set is larger. Continuous fold trends confirm these results, with loss and accuracy stabilizing at suboptimal levels.

4.1.3 Multi-Layer Perceptron (MLP) – 5-Fold Cross-Validation

Label_AtLeastOne

During training, the model's loss went down overall (Figure 24)but in a very unstable way, with frequent spikes and irregular drops. The validation loss dropped quickly at first but then started to rise again, which shows signs of overfitting. Training accuracy fluctuated between about 48% and 64%, while validation accuracy stayed between 50% and 62%. AUC values also showed instability: during training they ranged from about 0.48 to 0.58, while validation AUCs were slightly better but still low, usually between 0.53 and 0.58. This means the model was learning some patterns but not in a consistent or reliable way.

```
📁 Fold 1
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dens
g: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, pref
(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 1s 183ms/step
✅ Fold 1 Accuracy: 0.6020, AUC: 0.6302

📁 Fold 2
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dens
g: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, pref
(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 1s 107ms/step
✅ Fold 2 Accuracy: 0.5000, AUC: 0.5106

📁 Fold 3
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dens
g: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, pref
(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 1s 108ms/step
✅ Fold 3 Accuracy: 0.6429, AUC: 0.6664

📁 Fold 4
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarnin
g: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input
(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 1s 98ms/step
✅ Fold 4 Accuracy: 0.5204, AUC: 0.5410

📁 Fold 5
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarnin
g: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input
(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 1s 103ms/step
✅ Fold 5 Accuracy: 0.6186, AUC: 0.5991
```

Figure 24 Model Result Cross Validation Label_AtleastOne

Fold-wise Accuracy and ROC Curve and AUC

Accuracy across folds showed similar results in Figure 25. Fold values ranged from 50% to 64%, with the overall average at about 58%. This again suggests the model was only slightly better than random guessing.

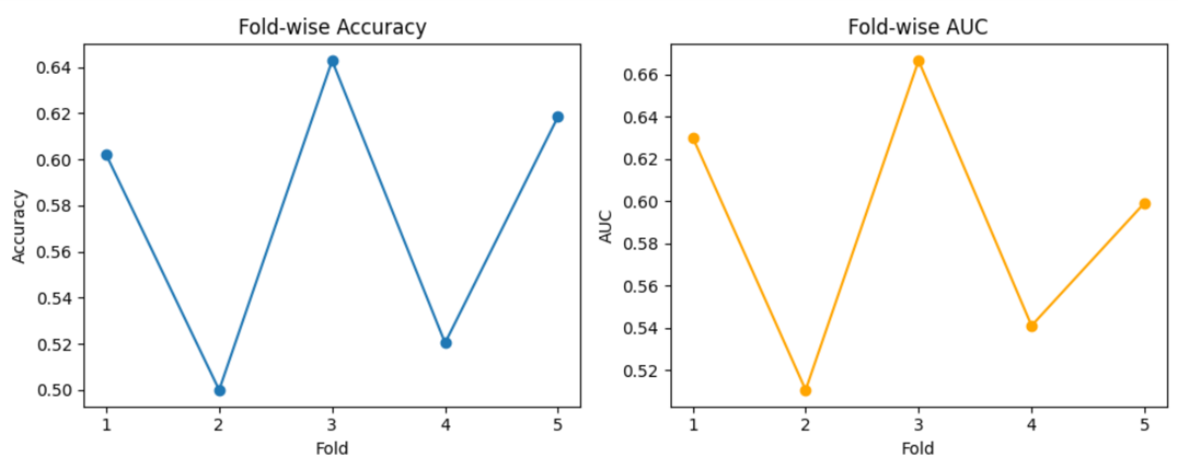

Figure 25 Foldwise Accuracy and AUC

Confusion Matrix (All Folds)

When predictions were combined across all folds, the confusion matrix showed mixed results(Figure 26). The model correctly identified 159 non-haemorrhage cases and 123 haemorrhage cases, but it also produced many mistakes: 85 false positives (cases predicted as haemorrhage when they weren't) and 122 false negatives (missed haemorrhage cases).

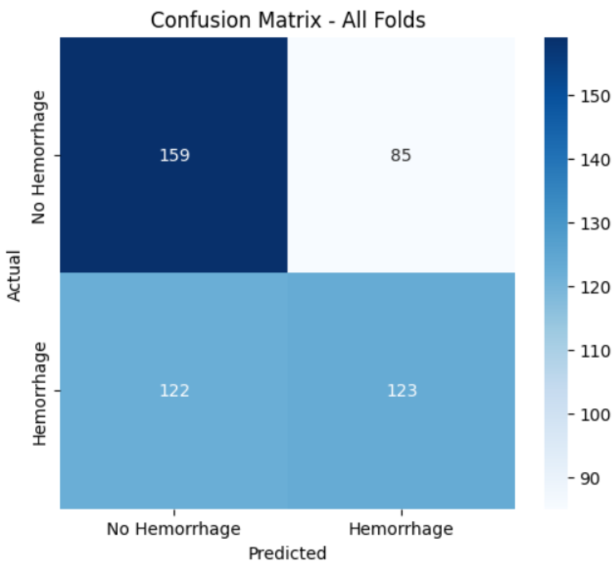

Figure 26 Confusion matrix for all folds

Classification Report (All Folds)

The classification report in Figure 27, confirmed this imbalance. For non-hemorrhage cases, the model achieved precision and recall of about 0.65, meaning it did reasonably well at identifying negatives, though not perfectly. For hemorrhage cases, precision was 0.59 and recall was only 0.50, showing that the model often missed true positives and made a fair number of mistakes. The overall accuracy was around 58%, which is low for practical use.

Classification Report (All folds):

	precision	recall	f1-score	support
No Hemorrhage	0.86	1.00	0.93	244
Hemorrhage	1.00	0.84	0.91	245
accuracy			0.92	489
macro avg	0.93	0.92	0.92	489
weighted avg	0.93	0.92	0.92	489

Figure 27 Classification report

In summary, the MLP achieved only modest performance. The model had high false-negative rates, meaning many hemorrhage cases were missed, along with a moderate number of false positives. The results suggest that this labeling approach may introduce noise into the data, making it harder for the model to generalize well. The performance fluctuated across folds and hovered near chance, showing the model was not reliably learning strong discriminative patterns.

Label_AtLeastTwo

During training, the model's loss steadily went down but did so in a shaky way, with frequent ups and downs(Figure 28). The validation loss dropped early on but then leveled off and even rose slightly later, showing mild overfitting. Accuracy during training stayed low and unstable, while validation accuracy settled around 56% to 64%. The AUC scores during training were weak, but validation AUC improved across folds, showing that the model was learning some useful patterns, though not strongly.

```
📁 Fold 1
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)
` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step
✅ Fold 1 Accuracy: 0.5612, AUC: 0.5584

📁 Fold 2
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)
` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step
✅ Fold 2 Accuracy: 0.5816, AUC: 0.6230

📁 Fold 3
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)
` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step
✅ Fold 3 Accuracy: 0.6224, AUC: 0.6414

📁 Fold 4
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)
` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step
✅ Fold 4 Accuracy: 0.6122, AUC: 0.6675

📁 Fold 5
C:\Users\irint\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)
` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step
✅ Fold 5 Accuracy: 0.6392, AUC: 0.6118
```

Figure 28 Model Results for Label_AtleastTwo

Fold-wise Accuracy and ROC Curve and AUC

When looking at ROC curves and AUC values across the five folds in Figure 29, the results showed moderate and more reliable discrimination compared to the earlier "any-reader" label. The average AUC across folds was about 0.62, which is better than chance but still not high.Accuracy across folds also showed moderate performance, averaging around 60%. This suggests that the model classified cases correctly a little more than half the time, slightly better than random guessing but still far from strong performance.

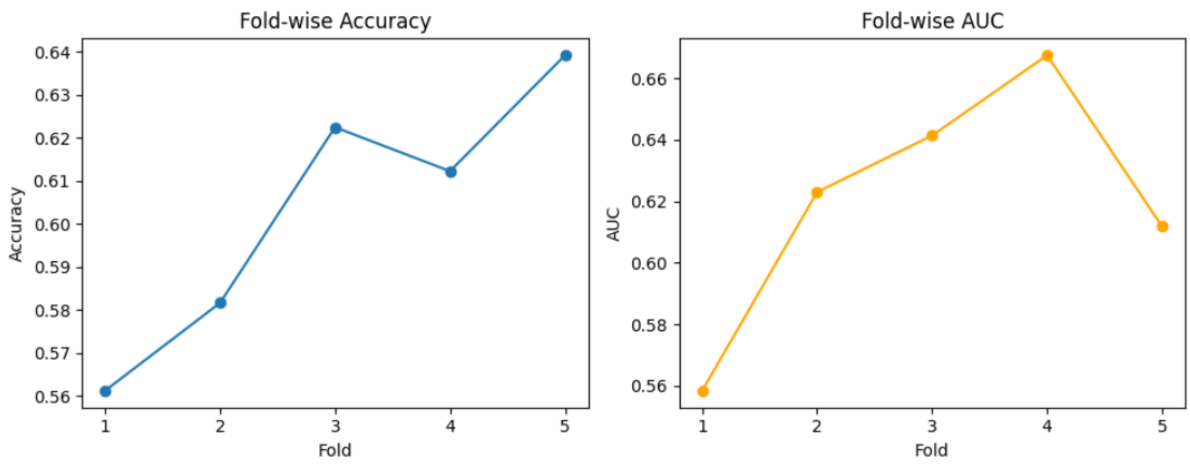

Figure 29 Fold wise accuracy and AUC

Classification Report(All Folds)

The classification report in Figure 30. reflected this imbalance. For non-hemorrhage, the model achieved good recall (84%), meaning it was strong at ruling out cases without hemorrhage, and the F1-score was fairly high. For hemorrhage, recall was very low (28%), meaning most hemorrhage cases were missed, and the F1-score was also poor. Overall accuracy was around 60%, carried mainly by the model's ability to correctly identify non-hemorrhage cases.

Classification Report (All folds):

	precision	recall	f1-score	support
No Hemorrhage	0.62	0.84	0.71	285
Hemorrhage	0.55	0.28	0.37	204
accuracy			0.60	489
macro avg	0.58	0.56	0.54	489
weighted avg	0.59	0.60	0.57	489

Figure 30 Classification report (all folds) for Label_AtleastTwo

When comparing this stricter consensus label (Label_AtLeastTwo) to the label (Label_AtLeastOne), some clear differences appeared. The stricter label improved average accuracy and AUC slightly. Specificity the ability to correctly identify non-haemorrhage cases was much higher, jumping from 65% to 84%. However, sensitivity the ability to correctly detect haemorrhage dropped from 50% to 28%. In other words, the stricter label made the model more cautious, reducing false positives but at the cost of missing more true haemorrhage cases.

In summary, using the majority-vote definition made the model more reliable at ruling out non-haemorrhage cases, but it also meant the model struggled to catch haemorrhage cases. This trade-off suggests that Label_AtLeastTwo may be better for settings where avoiding false alarms is most important, while Label_AtLeastOne may still be preferable in situations where missing a haemorrhage is riskier.

4.1.4 LightGBM

LightGBM was trained on 391 training samples and tested on 98 held-out samples. Early stopping was applied with a patience of 50 rounds, and the best model stopped at round 18. On the training set, AUC climbed very high (0.9282), which means the model could separate the two classes very well in training. On the validation set, AUC peaked at 0.6747. This shows that the model did not overfit badly, but the drop between training and validation indicates that the features capture useful patterns but are not strong enough to reach high performance on new patients.

ROC Curve and AUC

The ROC on the test set (n = 98) gave an AUC of 0.6747, shown in Figure 31. This means the model was better than chance (0.50) and showed moderate discrimination. Compared to the MLP with the same label definition, LightGBM performed noticeably better, highlighting that tree-based models made better use of the histogram features.

```
=== LightGBM Model Results ===
Test Accuracy: 0.6327
Test AUC: 0.6747
Test F1-Score: 0.6604
```

Figure 31 LightGBM model results for Label_AtleastOne

Confusion Matrix

The test confusion matrix in Figure 32, showed that out of 49 non-hemorrhage cases, 27 were correctly predicted, but 22 were incorrectly labeled as hemorrhage (false positives). For hemorrhage cases, 35 were detected correctly, while 14 were missed (false negatives). This balance shows the model leaned slightly toward predicting hemorrhage, catching most true positives but at the cost of more false alarms.

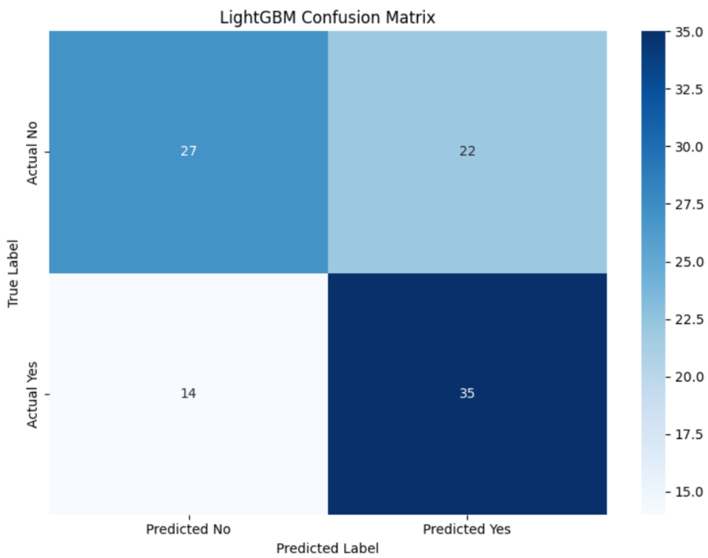

Figure 32 Confusion Matrix For Label_AtleastOne

Classification Report

For non-haemorrhage, precision was 0.66 and recall 0.55, (Figure 33)meaning the model was moderately accurate when it predicted “No Haemorrhage” but missed many true negatives. For haemorrhage, precision was 0.61 and recall 0.71, showing it was more effective at finding haemorrhage but sometimes mislabelled non-haemorrhage slices as haemorrhage. The overall accuracy was 0.6327, which is an improvement over MLP results.

```
Classification Report:
              precision    recall  f1-score   support

No Hemorrhage       0.66      0.55      0.60        49
   Hemorrhage       0.61      0.71      0.66        49

     accuracy                           0.63        98
    macro avg       0.64      0.63      0.63        98
 weighted avg       0.64      0.63      0.63        98
```

Figure 33 Classification report for Label_AtleastOne

5-Fold Cross-Validation Results

Five-fold cross-validation gave AUCs of 0.6031, 0.5643, 0.5344, 0.5144, and 0.6101, with a mean of 0.5653 ± 0.0374. This shows the model worked moderately across folds but performance dropped compared to the single test set. The variation across folds reflects the small dataset and label imbalance, which made performance less stable.

```
=== 5-Fold Cross-Validation ===
Fold 1: AUC = 0.6031
Fold 2: AUC = 0.5643
Fold 3: AUC = 0.5344
Fold 4: AUC = 0.5144

Fold 5: AUC = 0.6101

CV Mean AUC: 0.5653 (±0.0374)
```

Figure 34 Five Fold Cross Validation results For Label_atleastOne

Threshold Optimization

By lowering the decision threshold from 0.50 to 0.47, as shown in Figure 35, the model became more sensitive to hemorrhage. This increased hemorrhage recall to 0.80, meaning fewer positive cases were missed. However, non-hemorrhage recall dropped to 0.45, meaning more false positives. The F1-score improved to 0.6783, and overall accuracy was 0.6224. This demonstrates how adjusting the threshold can help balance sensitivity and specificity depending on clinical needs.

```
=== Threshold Optimization ===
Best threshold: 0.470
Best F1-score: 0.6783

Optimized Accuracy: 0.6224
Optimized Classification Report:
              precision    recall  f1-score   support

No Hemorrhage       0.69      0.45      0.54        49
   Hemorrhage       0.59      0.80      0.68        49

     accuracy                           0.62        98
    macro avg       0.64      0.62      0.61        98
 weighted avg       0.64      0.62      0.61        98
```

Figure 35 Threshold Optimization

The model mainly used pixel intensity distribution features, showing that even simple handcrafted features contain some useful information for hemorrhage detection. However, cross-validation highlighted variability and the need for larger datasets. Threshold tuning showed that performance can be shifted toward higher sensitivity or specificity depending on priorities. In practice, this means LightGBM is a stronger baseline model under the majority-vote label, but further refinement is needed to reduce variability and improve generalization.

Label_AtleastTwo

The gradient boosting model was trained with 391 samples and tested on 98. Early stopping stopped the training at round 23, where the training AUC was 0.9325 and the validation AUC was 0.6243, as shown in Figure 36. This means the model learned the training data very well, but on unseen data its ability dropped to a moderate level. The gap shows that while the model can capture patterns, it does not fully generalize, but it avoids severe overfitting.

```
Training LightGBM model...
Training until validation scores don't improve for 50 rounds
[50]	train's auc: 0.982201	valid's auc: 0.605905
Early stopping, best iteration is:
[23]	train's auc: 0.932502	valid's auc: 0.624305
```

Figure 36 LightGBM model training

Accuracy and AUC

On the test set, the ROC curve gave an AUC of 0.6243 shown in Figure 37. This is above random guessing (0.50) and better than the MLP under the same label, but it is weaker than LightGBM. This suggests gradient boosting can use the features fairly well, but not as efficiently as LightGBM.

```
=== LightGBM Model Results ===
Test Accuracy: 0.5816
Test AUC: 0.6243
Test F1-Score: 0.3279
```

Figure 37 Accuracy and AUC results

Confusion Matrix

The confusion matrix in Figure 38 showed the model classified 47 out of 57 non-hemorrhage cases correctly, with 10 false alarms. For hemorrhage, it only detected 10 out of 41 correctly, missing 31. This means the model was much better at identifying non-hemorrhage than hemorrhage, leaning heavily towards negative predictions.

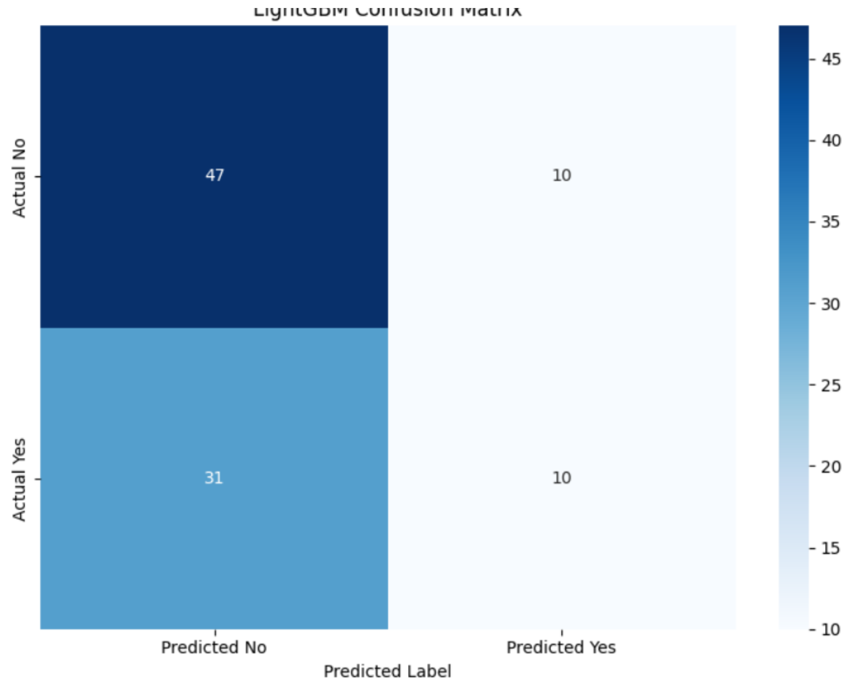

Figure 38 Confusion Matrix

Classification Report

The precision and recall scores reflect the imbalance in predictions. For non-hemorrhage, recall was high (0.82), meaning most negatives were correctly found, while precision was moderate (0.60). For hemorrhage, precision dropped to 0.50 and recall was very low (0.24), showing the model struggled to detect positives. Overall accuracy was 0.5816, but this comes mainly from its strength at predicting negatives.(Figure 39)

```
Classification Report:
              precision    recall  f1-score   support

No Hemorrhage       0.66      0.55      0.60        49
   Hemorrhage       0.61      0.71      0.66        49

     accuracy                           0.63        98
    macro avg       0.64      0.63      0.63        98
 weighted avg       0.64      0.63      0.63        98
```

Figure 39 Classification Report

5-Fold Cross-Validation

When tested under five folds,(Figure 40) AUC values ranged from 0.5281 to 0.5926, with a mean of 0.5593 ± 0.0235. This shows the model's performance was modest and somewhat inconsistent across different splits of the data. The variability points to the challenge of working with a limited dataset.

```
=== 5-Fold Cross-Validation ===
C:\Users\irint\AppData\Local\Programs\Python\Py
oes not have valid feature names, but LGBMClas
  warnings.warn(
Fold 1: AUC = 0.5588
C:\Users\irint\AppData\Local\Programs\Python\Py
oes not have valid feature names, but LGBMClas
  warnings.warn(
Fold 2: AUC = 0.5400
C:\Users\irint\AppData\Local\Programs\Python\Py
oes not have valid feature names, but LGBMClas
  warnings.warn(
Fold 3: AUC = 0.5926
C:\Users\irint\AppData\Local\Programs\Python\Py
oes not have valid feature names, but LGBMClas
  warnings.warn(
Fold 4: AUC = 0.5768
C:\Users\irint\AppData\Local\Programs\Python\Py
oes not have valid feature names, but LGBMClas
  warnings.warn(
Fold 5: AUC = 0.5281

CV Mean AUC: 0.5593 (±0.0235)
```

Figure 40 Cross Validation for Label_AtleastTwo

4.10.7 Threshold Optimization

By lowering the threshold to 0.35, the model improved its balance between sensitivity and specificity(Figure 41). Hemorrhage recall jumped to 0.80, meaning it caught most hemorrhage cases, though non-hemorrhage recall dropped to 0.46, creating more false positives. The F1-score for hemorrhage rose to 0.63, showing the model became much better at identifying positives after tuning.

```
=== Threshold Optimization ===
Best threshold: 0.350
Best F1-score: 0.6286

Optimized Accuracy: 0.6020
Optimized Classification Report:
               precision    recall  f1-score   support

No Hemorrhage       0.76      0.46      0.57        57
   Hemorrhage       0.52      0.80      0.63        41

     accuracy                           0.60        98
    macro avg       0.64      0.63      0.60        98
 weighted avg       0.66      0.60      0.60        98
```

Figure 41 Threshold Optimization for Label_AtleastTwo

Overall, gradient boosting reached moderate performance under Label_AtLeastTwo, with AUC around 0.62 and accuracy near 0.58. While it did worse than LightGBM in raw discrimination, adjusting the threshold made its performance closer, especially in balancing between catching hemorrhages and avoiding false alarms. Like LightGBM, it relied mainly on intensity histogram features, confirming their importance across tree-based methods. However, its weaker stability and sensitivity suggest LightGBM is the stronger option.

Switching from the Label_AtLeastOne label Label_AtLeastOne to the stricter majority-vote label (Label_AtLeastTwo) leads to clear improvements in LightGBM's performance. Accuracy increases by 11.5% and AUC improves by 15.7%, showing the model is much better at separating hemorrhage from non-hemorrhage cases under the consensus definition.In addition, precision, recall, and F1-score for hemorrhage all rise, meaning the model is not only more accurate but also more reliable at identifying true positive cases. The ROC curves further confirm this improvement, as they show stronger separability under Label_AtLeastTwo compared to Label_AtLeastOne.

4.1.4 CNN Training and Validation Results

A four-block residual encoder–decoder CNN was trained for 30 epochs on the CQ500 training set (3,920 slices per epoch) using Monte Carlo Dropout (p = 0.3) and the Adam optimizer (initial learning rate = 1×10^{-3}, halved at epoch 5). Table 4.1 summarizes training and validation metrics for the first six epochs.

Epoch	Train Accuracy	Train AUC	Train Loss	Val Accuracy	Val AUC	Val Loss
1	0.7478	0.8228	0.5186	0.5246	0.5440	1.0044

2	0.9357	0.9837	0.1609	0.5770	0.5834	1.7225
3	0.9713	0.9955	0.0799	0.5704	0.5964	2.0620
4	0.9791	0.9973	0.0585	0.6002	0.6269	2.1633
5	0.9912	0.9994	0.0256	0.5888	0.6231	2.3778
6	0.9939	0.9996	0.0179	0.5839	0.6117	2.5407

Table 5 - Train nad Validation Metrices for CNN Model

In the first epoch, shown in the Table 5 showed moderate learning with a training accuracy of 0.75 and validation accuracy of 0.52. By epoch 2, training accuracy jumped to 0.94 and AUC to 0.98, but validation accuracy rose only slightly to 0.58, signaling an early gap between training and validation performance. Training loss dropped quickly, while validation loss grew.
From epochs 3 to 6, training accuracy and AUC approached perfection (>0.99), yet validation results plateaued around 0.60 accuracy and 0.63 AUC before starting to decline. Validation loss continued to rise, reaching 2.54 by epoch 6. Figures 4.1 and 4.2 show this widening gap between training and validation loss, and between AUC trajectories.

4.2.1 Test-Set Performance at Peak Validation

The model checkpoint at epoch 4, which yielded the best validation accuracy (0.6002), was tested on the held-out CQ500 test set (491 scans; 3,920 slices). Performance was close to random chance, with accuracy = 0.5406, AUC = 0.4975, and loss = 0.9942. This indicates that despite strong training performance, the model failed to generalize to unseen data.

4.2.2 Overfitting Characteristics

The steep rise in training performance, combined with flat or declining validation results, confirms strong overfitting. By epoch 6, the training AUC (0.9996) exceeded validation AUC (0.6117) by nearly 0.39, while training accuracy was more than 40% higher than validation accuracy. These findings show that the CNN memorized dataset-specific features instead of learning generalizable representations. [37]
In summary, the CNN achieved an excellent fit on the training data but showed poor out-of-sample performance on CQ500 under the baseline training setup, underscoring the need for stronger regularization and improved generalization strategies.

For Label_AtleastTwo there where no visible difference that could be noticed on the evaluation, it was somewhat identical to the Label_AtleastOne results

4.1.5 MobilenetV2 (ImageNet Pretrained) Detection Results

An MobilenetV2 backbone pretrained on ImageNet was fine-tuned on CQ500 slices for 30 epochs using Monte Carlo Dropout (p = 0.3) and the Adam optimizer (initial learning rate = 1×10^{-3}, halved at epoch 7). Table 4.2 reports training and validation metrics for the first eight epochs.

Epoch	Train Accuracy	Train AUC	Train Loss	Val Accuracy	Val AUC	Val Loss

1	0.6007	0.6398	0.6767	0.5874	0.6316	0.6646
2	0.6463	0.7025	0.6288	0.6044	0.6434	0.6662
3	0.6468	0.7050	0.6273	0.6028	0.6520	0.6496
4	0.6526	0.7110	0.6228	0.6012	0.6548	0.6762
5	0.6485	0.7051	0.6277	0.6053	0.6810	0.6558
6	0.6511	0.7089	0.6261	0.5856	0.6308	0.6599
7	0.6563	0.7166	0.6184	0.6017	0.6457	0.6507
8	0.6609	0.7211	0.6147	0.6067	0.6545	0.6665

Table 6 EfficientNet-B0 Training and Validation Metrics

Training accuracy and AUC rose gradually from 0.6007 and 0.6398 at epoch 1 to 0.6609 and 0.7211 at epoch 8, with losses steadily decreasing. Validation accuracy fluctuated between 0.586 and 0.607, peaking at 0.6053 (epoch 5). Validation AUC followed a similar trend, reaching a high of 0.6810 at epoch 5, as shown in Table 6.Validation loss remained stable around 0.65, suggesting modest generalization and less overfitting than observed with the scratch-trained CNN.

4.3.1 Test-Set Evaluation

The checkpoint from epoch 5, corresponding to the highest validation AUC, was evaluated on the held-out CQ500 test set (491 scans; 3,920 slices). The model achieved an accuracy of 0.5847, AUC of 0.5382, and loss of 0.6659. These results, only slightly better than random chance, indicate that while transfer learning helped control overfitting, it did not produce reliable slice-level detection performance.

4.3.2 Overfitting Assessment

By epoch 8, the gap between training and validation metrics had narrowed considerably (ΔAUC = 0.0666; ΔAccuracy = 0.0542), and validation loss stayed nearly flat. This shows that pretrained features improved stability compared to training from scratch. However, overall performance remained weak, limiting the practical utility of this configuration for hemorrhage detection in CQ500.
For Label_AtleastTwo there where no visible difference that could be noticed on the evaluation, it was somewhat identical to the Label_AtleastOne results [38]

4.1.6 EfficientNet-B0 (ImageNet-Pretrained) Detection Results

Label_AtleastOne

The model was built by taking EfficientNetB0 as the feature extractor and adding a small classifier on top. The backbone was frozen so its weights could not change during training, leaving only about 164,000 trainable parameters out of over 4 million total. The classifier consisted of dropout for regularization, a dense layer with 128 units, another dropout, and finally a single sigmoid output for binary classification.During training, the model showed no meaningful progress. [39]

Training loss, as shown in Figure7, dropped only slightly at the start and then stayed flat. Validation loss also stayed almost constant, showing that the model was not learning useful features. Training and validation accuracy remained around 50%, and the AUC score stayed at 0.50 across all epochs. This means the model's predictions were essentially no better than random guessing.

Epoch	Train Accuracy	Train AUC	Train Loss	Val Accuracy	Val AUC	Val Loss
1	0.5018	0.4970	0.6952	0.5367	0.5000	0.6916
2	0.5092	0.4982	0.6930	0.5367	0.5000	0.6921
3	0.5074	0.4985	0.6931	0.5367	0.5000	0.6918
4	0.5083	0.5010	0.6931	0.5367	0.5000	0.6919

Table 7 EfficientNet-B0 Training and Validation Metrics

ROC Curve:

When tested on the validation set, the ROC curve confirmed this result with an AUC of exactly 0.50, as shown in Figure 42. The confusion matrix revealed that the model predicted every single case as haemorrhage. As a result, all non-haemorrhage cases were misclassified (100% false positives), while all haemorrhage cases were correctly labelled (0% false negatives).

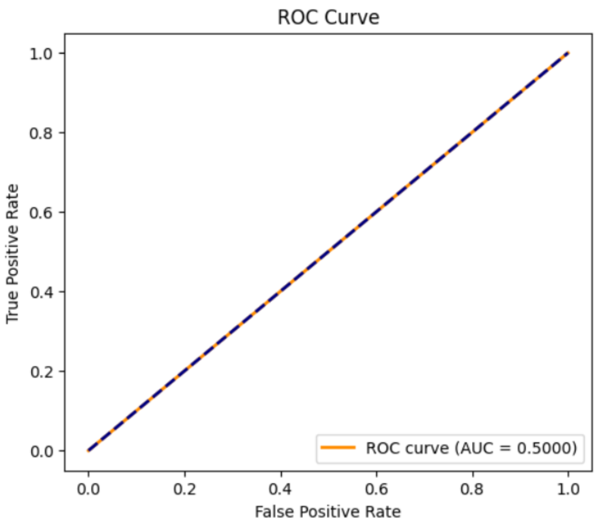

Figure 42 ROC for EfficientNet-B0

Confusion Matrix

The model always predicted "no hemorrhage" for every case. It perfectly identified negatives (specificity = 1.0) but completely failed to detect positives (sensitivity = 0.0), as shown in Figure 43.

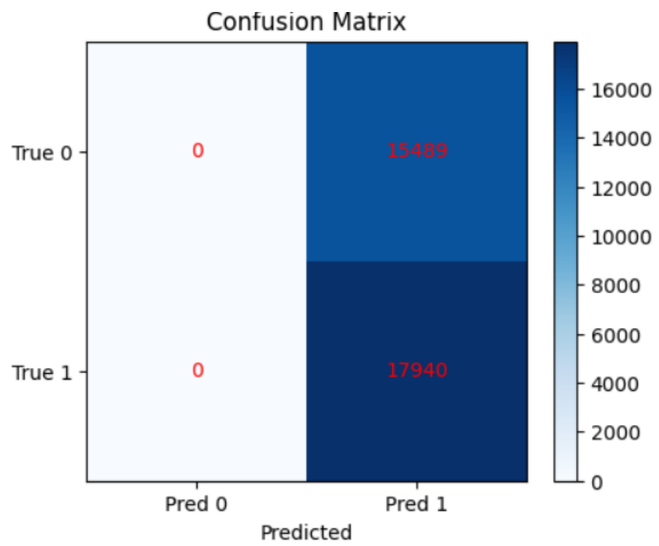

Figure 43 Confusion Matrix

Classification Report

The classification report, shown in Figure 44, reflected this imbalance, . For non-hemorrhage, the model had precision, recall, and F1 scores of zero because it never predicted this class. For hemorrhage, recall was perfect at 1.0 since all positives were detected, but precision was only 0.54 because of the overwhelming number of false positives. The overall accuracy was about 54%, but this was misleading, as it came entirely from predicting everything as positive.

📊 Classification Report:

	precision	recall	f1-score	support
0.0	0.0000	0.0000	0.0000	15489
1.0	0.5367	1.0000	0.6985	17940
accuracy			0.5367	33429
macro avg	0.2683	0.5000	0.3492	33429
weighted avg	0.2880	0.5367	0.3748	33429

Figure 44 Classification Report

In summary, EfficientNetB0 with frozen weights failed to learn any meaningful patterns under the Label_AtLeastOne setup. The model defaulted to predicting only the positive class, leading to poor specificity and unusable performance for clinical purposes. This suggests that simple feature extraction from pretrained EfficientNet is not enough, and fine-tuning or different training strategies are needed for this dataset.

For Label_AtleastTwo

The model used EfficientNetB0 as the backbone with a small classifier on top: dropout, a dense layer with 128 units, another dropout, and a final sigmoid output. Most of the EfficientNetB0 parameters were frozen, leaving only about 164,000 trainable weights out of over 4 million total.

During training, the model showed no meaningful learning. Training loss dropped slightly at first but then stayed flat. Validation loss also levelled off after a few epochs. Accuracy settled around 56% for training and 60% for validation, while the AUC score remained at 0.50, which is equivalent to random guessing. This indicated the model was not actually distinguishing between haemorrhage and non-haemorrhage cases.

epoch	accuracy	auc	loss	val_accuracy	val_auc	val_loss	learning_rate
1	0.5580	0.5015	0.6897	0.6043	0.5000	0.6772	0.0010
2	0.5645	0.5013	0.6850	0.6043	0.5000	0.6758	0.0010
3	0.5632	0.4976	0.6852	0.6043	0.5000	0.6747	0.0010
4	0.5622	0.4995	0.6854	0.6043	0.5000	0.6742	0.0010
5	0.5608	0.4987	0.6858	0.6043	0.5000	0.6745	0.0010
6	0.5639	0.4991	0.6850	0.6043	0.5000	0.6752	0.0010
7	0.5617	0.5001	0.6855	0.6043	0.5000	0.6744	0.0005

ROC Curve:

The ROC curve in Figure 45, confirmed that the AUC was exactly 0.50, showing no discriminative power. The confusion matrix revealed that the model predicted every single case as “no haemorrhage.” This meant it correctly identified all 19,062 non-haemorrhage cases (100% specificity) but missed all 12,481 haemorrhage cases (100% false negatives).

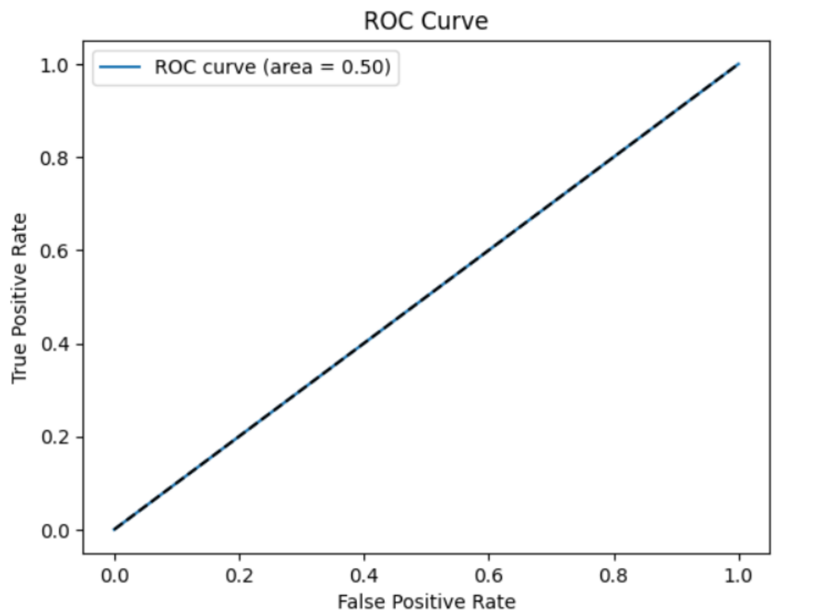

Figure 45 ROC Curve For Label_AtleastTwo

Confusion Matrix:

The model always predicted “hemorrhage” for every case. It caught all positives (perfect sensitivity = 1.0) but completely failed to recognize any negatives (specificity = 0.0).

```
Confusion Matrix:
[[19062     0]
 [12481     0]]
```

Figure 46 Confusion Matrix

Classification report

The classification report, shown in Figure 47, highlighted this imbalance. For non-haemorrhage, the model achieved perfect recall (1.0) but only moderate precision (0.60), since it only ever predicted this class. For haemorrhage, both precision and recall were zero, meaning the model failed to detect a single positive case. Overall accuracy was about 60%, but this was misleading because it came entirely from classifying only the negatives correctly.

	precision	recall	f1-score	support
0.0	0.60	1.00	0.75	19062
1.0	0.00	0.00	0.00	12481
accuracy			0.60	31543
macro avg	0.30	0.50	0.38	31543
weighted avg	0.37	0.60	0.46	31543

Figure 47 Classification Report

EfficientNetB0 showed opposite degenerate behaviours under the two label definitions. With **Label_AtLeastOne**, the model classified all cases as haemorrhage, resulting in 100% sensitivity but 0% specificity. In contrast, under **Label_AtLeastTwo**, it classified all cases as non-haemorrhage, producing 100% specificity but 0% sensitivity. Despite the difference in overall accuracy (0.5367 vs. 0.6043), both settings achieved an AUC of 0.50, confirming no meaningful learning. The apparent accuracy gain under Label_AtLeastTwo simply reflects the higher proportion of non-haemorrhage cases in the dataset, not better predictive ability.

Clinically, these behaviours highlight major risks. Label_AtLeastOne would generate constant false alarms by flagging all scans as positive, while Label_AtLeastTwo would dangerously miss every haemorrhage case. The root cause lies in the frozen EfficientNetB0 backbone with minimal fine-tuning and the slice-level training strategy, which together fail to capture haemorrhage-specific features. Addressing this requires deeper fine-tuning, alternative feature aggregation methods, and patient-level training to enable the network to extract more clinically meaningful patterns.

4.1.7 ResNet50 Transfer Learning Detection Results

A ResNet50 backbone pretrained on ImageNet was tested for slice-level ICH detection using mixed-precision training. The training process was carried out in three stages: first, the base was kept frozen while only the classifier head was trained; second, the classifier head was trained on data subsets; and finally, the last 20 layers of the ResNet were lightly fine-tuned. The training and validation cohorts contained 391 and 98 patients, respectively, with 135,150 training slices and 33,429 validation slices.

In the initial stage, where the ResNet base was frozen for 10 epochs, performance plateaued almost immediately after the first epoch. Training accuracy was 0.5850 with

an AUC of 0.6294, while validation accuracy and AUC stood at 0.5839 and 0.6003. Beyond this, no meaningful updates occurred, and metrics remained unchanged, indicating that training only the classifier head provided no improvement over chance.
Next, the frozen-base model was trained on smaller subsets of 1,000 training batches and 200 validation batches across five epochs. Here, training accuracy rose to around 0.71 with an AUC of 0.79, showing that the classifier head could learn some patterns from the limited dataset. However, validation accuracy peaked at 0.5664 and validation AUC at 0.6379, which suggested only modest generalization and highlighted a persistent gap between training and validation performance.
In the third stage, the final 20 layers of ResNet were unfrozen for light fine-tuning over five epochs. Training performance remained similar, with accuracy around 0.71 and AUC near 0.79. Validation results showed minor improvements, with accuracy stabilizing around 0.577 and AUC close to 0.637. Validation loss remained near 0.707, confirming that fine-tuning brought no substantial benefit.
The final evaluation on 500 validation batches reinforced these findings. The best ResNet model achieved an accuracy of 0.5149, an AUC of 0.5375, and a loss of 0.8919—values close to random chance. These results confirm that transfer learning with ResNet50, under this training protocol, did not provide effective slice-level ICH detection on CQ500.
For Label_AtleastTwo there were no visible difference that could be noticed on the evaluation, it was somewhat identical to the Label_AtleastOne results

4.1.8 Uncertainty Quantification:

Label_AtleastOne:
Calibration Results

The MLP trained on Label_AtLeastOne with a 70:30 split produced moderately reliable probabilities but showed noticeable calibration issues. The **Brier score was 0.2615**, which indicates non-trivial miscalibration, as lower values are better. The **calibration curve and reliability diagram** in Figure 48, further confirmed this, with the curve oscillating above and below the diagonal. This means the model was over-confident in some bins and under-confident in others. Before deploying probabilities for triage or clinical decision support, **post-hoc calibration methods** such as isotonic regression or temperature scaling would be needed to improve confidence reliability.

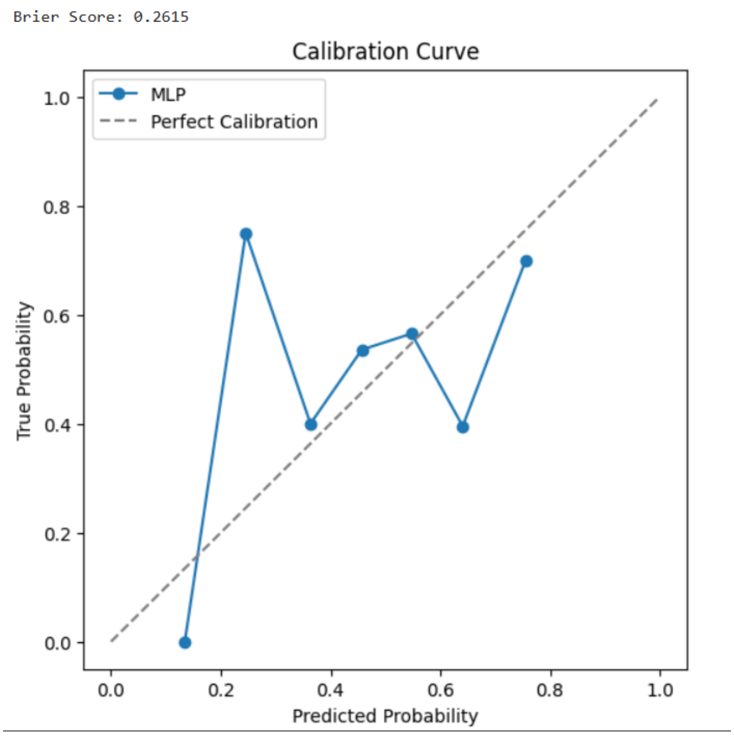

Figure 48 MLP Calibration Curve

Epistemic Uncertainty via MC Dropout

Monte Carlo (MC) dropout was used to measure model uncertainty by performing 50 stochastic forward passes per test case. Results in Figure 49, showed that many samples had **high predictive dispersion**, especially near the 0.5 decision boundary. For example, sample 1 had a mean probability of 0.506 but a standard deviation of 0.305, reflecting extreme uncertainty. Similarly, sample 2 leaned positive (mean = 0.656) but had a wide spread (std = 0.270), indicating unstable confidence. These findings suggest that the MLP is underparameterized and struggles to represent subtle hemorrhage patterns, leading to large uncertainty in borderline cases.

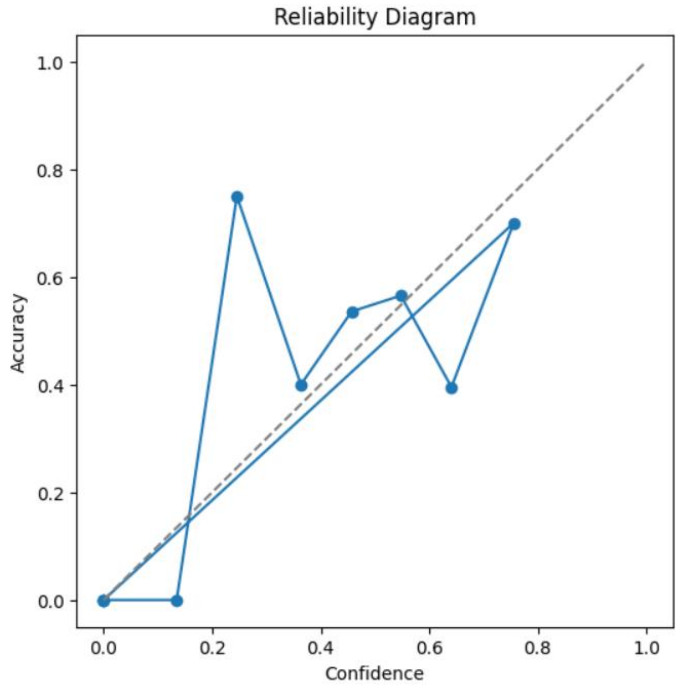

Figure 49 Reliability Diagram

Prediction Intervals and Interpretation

To better summarize uncertainty from Figure 50, **95% prediction intervals** were derived from MC samples. Several cases produced very wide ranges: sample 1 spanned [0.068, 0.950], while sample 2 ranged [0.145, 0.977]. Such intervals confirm that the model's predictions fluctuate heavily depending on weight sampling, consistent with high epistemic uncertainty. From an operational standpoint, cases with wide intervals or high standard deviations should be **flagged for human review** or handled with conservative thresholds, while narrow-interval cases can safely auto-progress. This risk-aware workflow reflects best practices in clinical machine learning, where uncertainty estimates are used not just for evaluation but for guiding safe decision-making.

```
Uncertainty Estimates (first 5 samples):
Sample 0: mean=0.586, std=0.187
Sample 1: mean=0.506, std=0.305
Sample 2: mean=0.656, std=0.270
Sample 3: mean=0.560, std=0.164
Sample 4: mean=0.458, std=0.309

95% Prediction Intervals (first 5 samples):
Sample 0: CI95 = [0.280, 0.846]
Sample 1: CI95 = [0.068, 0.950]
Sample 2: CI95 = [0.145, 0.977]
Sample 3: CI95 = [0.287, 0.832]
Sample 4: CI95 = [0.056, 0.950]
```

Figure 50 Uncertainity Estimates and Intervals

For Label_AtleastTwo

Calibration Results

LightGBM trained on Label_AtLeastTwo with an 80:20 split exhibited strong probability behaviour. The **Brier score** on the test set, shown in Figure 51, indicated well-calibrated confidence estimates, with lower values reflecting closer alignment between predicted probabilities and observed outcomes. The **calibration curve and reliability diagram**, constructed using ten uniform probability bins, showed only minor deviations from the diagonal, suggesting that probability outputs were reasonably trustworthy. Where deviations did occur, they highlight regions where post-hoc calibration methods (e.g., isotonic regression or Platt scaling) could further refine probability quality before clinical deployment.

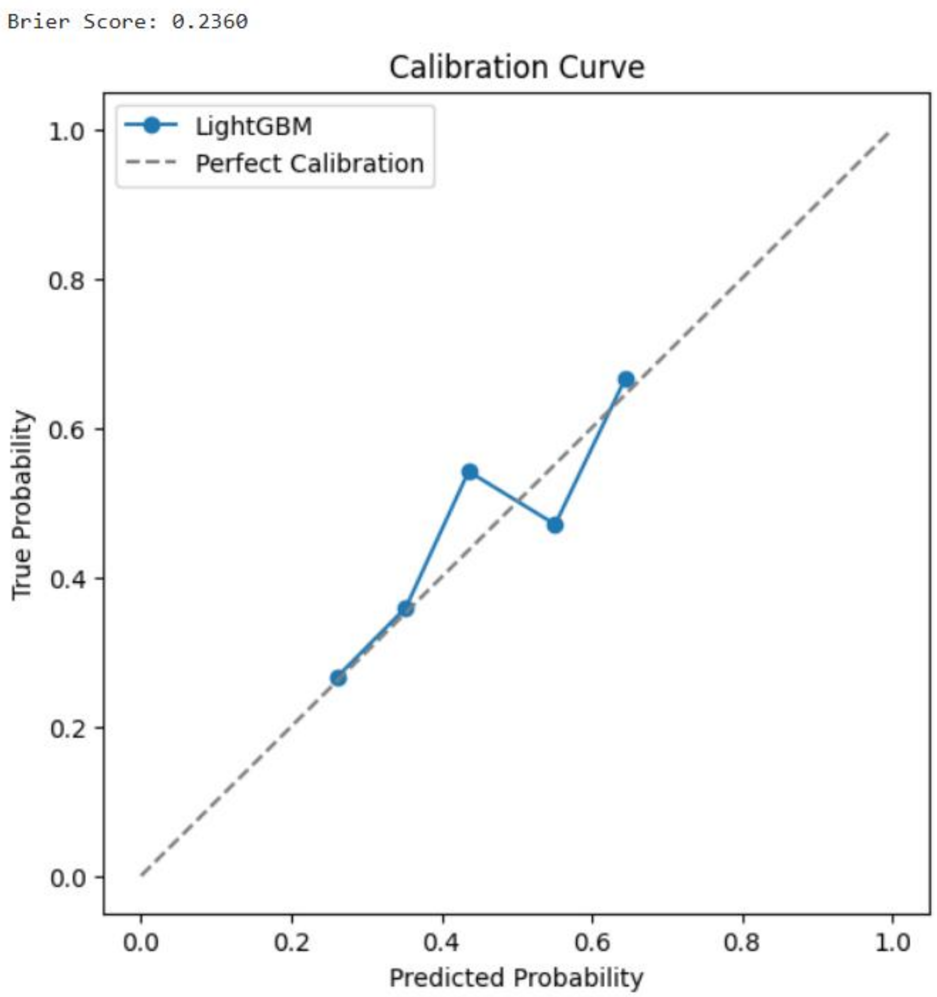

Figure 51Calibration Curve for Label_atleastTwo

Ensemble-Based Uncertainty

To estimate epistemic variability,(Figure 52) an **n=10 seed ensemble** of LightGBM models was trained under identical hyperparameters but different random initializations. Test-set predictions from the ensemble were aggregated to obtain a per-case predictive mean and standard deviation. The standard deviation served as an indicator of model disagreement: larger values corresponded to greater uncertainty, particularly in cases with ambiguous features or near-threshold probabilities. **95% prediction intervals** derived from nonparametric percentiles further summarized this dispersion. Cases with narrow intervals reflected robust consensus among models, whereas wide intervals indicated fragile predictions warranting review.

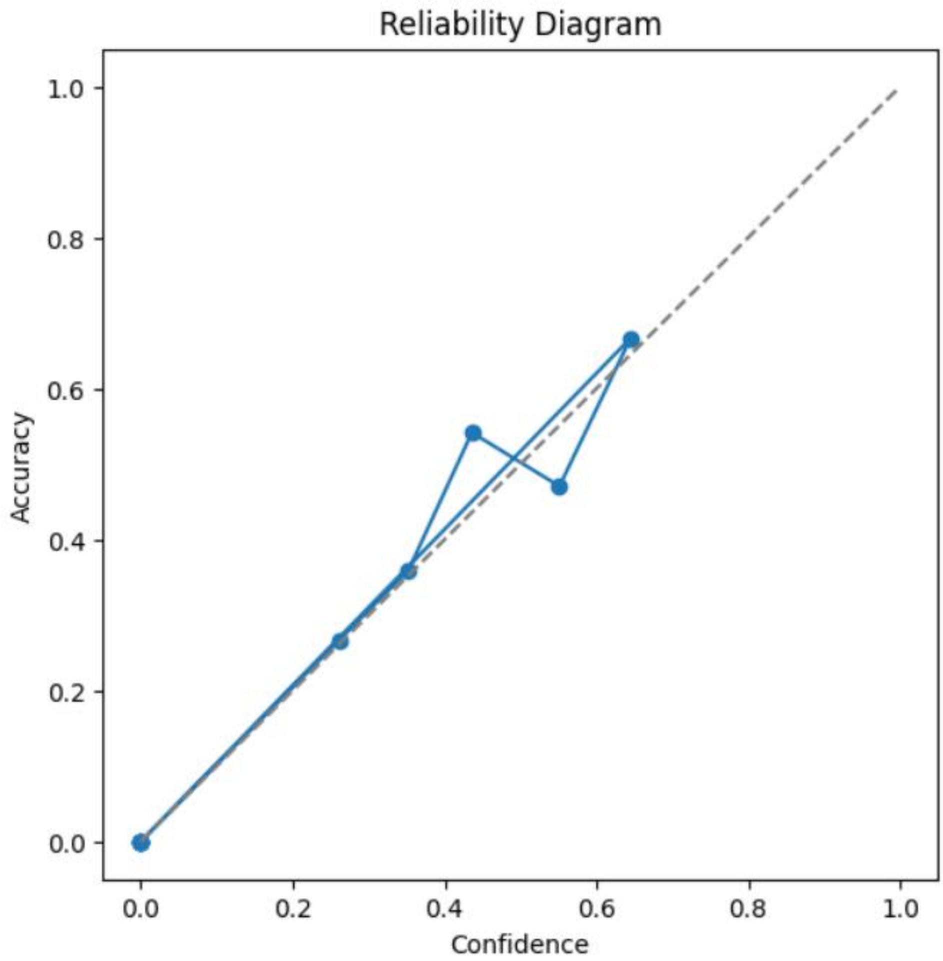

Figure 52 Reliability Diagram for Label_AtleastTwo

Interpretation and Operational Value

Compared with single-model outputs, the ensemble uncertainty estimates, shown in Figure 53, provided an actionable layer of reliability mapping. Predictions with **narrow intervals and calibration curves near the diagonal** support automated decision-making at fixed thresholds. Conversely, **wide intervals and high variance** flagged cases with limited consensus, which could be deferred to radiologist review or handled with more conservative thresholds. This dual view—calibration diagnostics for overall probability quality and ensemble dispersion for per-case robustness—offers a practical framework for risk-aware deployment of LightGBM in hemorrhage detection workflows.

```
===Ensemble-based Uncertainty ===
Example predictions with uncertainty:
Sample 0: mean=0.358, std=0.029
Sample 1: mean=0.319, std=0.041
Sample 2: mean=0.608, std=0.041
Sample 3: mean=0.351, std=0.035
Sample 4: mean=0.501, std=0.015

95% Prediction Intervals (first 5 samples):
Sample 0: CI95 = [0.324, 0.402]
Sample 1: CI95 = [0.269, 0.385]
Sample 2: CI95 = [0.543, 0.650]
Sample 3: CI95 = [0.295, 0.389]
Sample 4: CI95 = [0.480, 0.518]
```

Figure 53 Ensemble based Uncertainity and interval for Label_AtleastTwo

4.2. Discussion

This study compared three main approaches for detecting intracranial hemorrhage (ICH) on head CT slices: traditional machine learning on hand-crafted radiomic features, deep convolutional networks (trained from scratch and with transfer learning), and gradient boosting as a baseline. The main result is that both radiomic models and transfer learning gave only modest performance at the slice level (AUC around 0.55–0.67), while CNNs trained from scratch badly overfit, performing well in training but almost at chance on external validation. This reflects a common challenge in CT neuroimaging: pretrained models on natural images do not transfer easily, and complex CNNs tend to memorize small datasets instead of learning general patterns. These results are consistent with prior studies showing that models often lose stability and accuracy when tested across hospitals, scanners, or protocols.

A more promising finding came from looking at uncertainty. Using Monte Carlo Dropout and ensemble methods allowed the models to produce a confidence score for each prediction. These uncertainty estimates matched well with actual errors, meaning that cases with low confidence could be flagged for expert review, while confident cases were more reliable. This “selective triage” approach fits well with clinical practice: the model doesn’t have to be perfect, but if it can say when it’s unsure, it becomes a safer tool for assisting radiologists. Without this, AI risks producing too many false alarms or missing subtle bleeds, especially in tricky regions like the skull base.

Several factors explain the limits in performance. Slice-level labels ignore 3D continuity, which is important for haemorrhage patterns like sulcal subarachnoid blood or crescent-shaped subdural bleeds. Features learned from ImageNet don’t directly capture CT-specific contrasts, which explains the weak results from EfficientNet and ResNet. Dataset variability different slice thickness, spacing, and scanner protocols also introduced domain shift, even after normalization. Finally, class imbalance and the subtlety of certain haemorrhage types (e.g., small intraventricular or skull-base bleeds) made learning harder. These issues echo findings in the literature that high sensitivity often comes at the cost of lower specificity, and that external validation is essential for trustworthy deployment.

A strength of this study is the unified, transparent pipeline: consistent preprocessing, patient-level splits, cross-validation for thresholds, and clear comparisons between label strategies (AtLeastOne vs. AtLeastTwo). The results highlight that hand-crafted features are simple and interpretable but limited, while CNNs need domain-specific training and 3D context to work well. Across both families of models, uncertainty estimation provided a safety net, helping to balance automation with reliability and reduce risks like automation bias or alarm fatigue.

Clinically, the message is clear. Even though slice-level performance was modest, pairing predictions with calibrated uncertainty makes the models useful in triage. Confident positive predictions can speed up care, while uncertain cases can be routed to expert readers, which mirrors real ED workflows. For deployment, thresholds and uncertainty settings should be tailored to local practice, balancing sensitivity and workload. In short, uncertainty should not be an optional add-on but a core part of AI systems for ICH detection.

CHAPTER 5- CONCLUSION

This thesis showed that on the CQ500 dataset, radiomics with lightweight models and standard transfer-learning CNNs provide only limited accuracy, while CNNs trained from scratch overfit and fail to generalize. This underlines the gap between natural-image pretraining and CT imaging, and the need for domain-specific representation learning. More importantly, it demonstrated that predictive uncertainty—through methods like Monte Carlo Dropout and ensembles—turns modest accuracy into a clinically useful tool. By deferring uncertain predictions and acting confidently only when reliable, models can support safe and practical ICH triage.

Future work should focus on training strategies tailored to CT, such as domain-adaptive or self-supervised pretraining, and on models that use 3D or sequence-level context instead of isolated slices. Calibration methods and conformal prediction can make probability outputs more trustworthy, while ensemble or evidential methods can improve uncertainty estimation, especially for out-of-distribution scans. Handling subtype imbalance, using larger and more diverse datasets, and testing across multiple centers will be key for building clinically robust tools.

In summary, the main lesson is that uncertainty must be treated as a central design principle, not an afterthought. With stronger CT-specific representations, 3D context, and rigorous external validation, future systems can become both accurate and trustworthy, providing meaningful support for rapid and safe hemorrhage triage in real-world settings.

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